

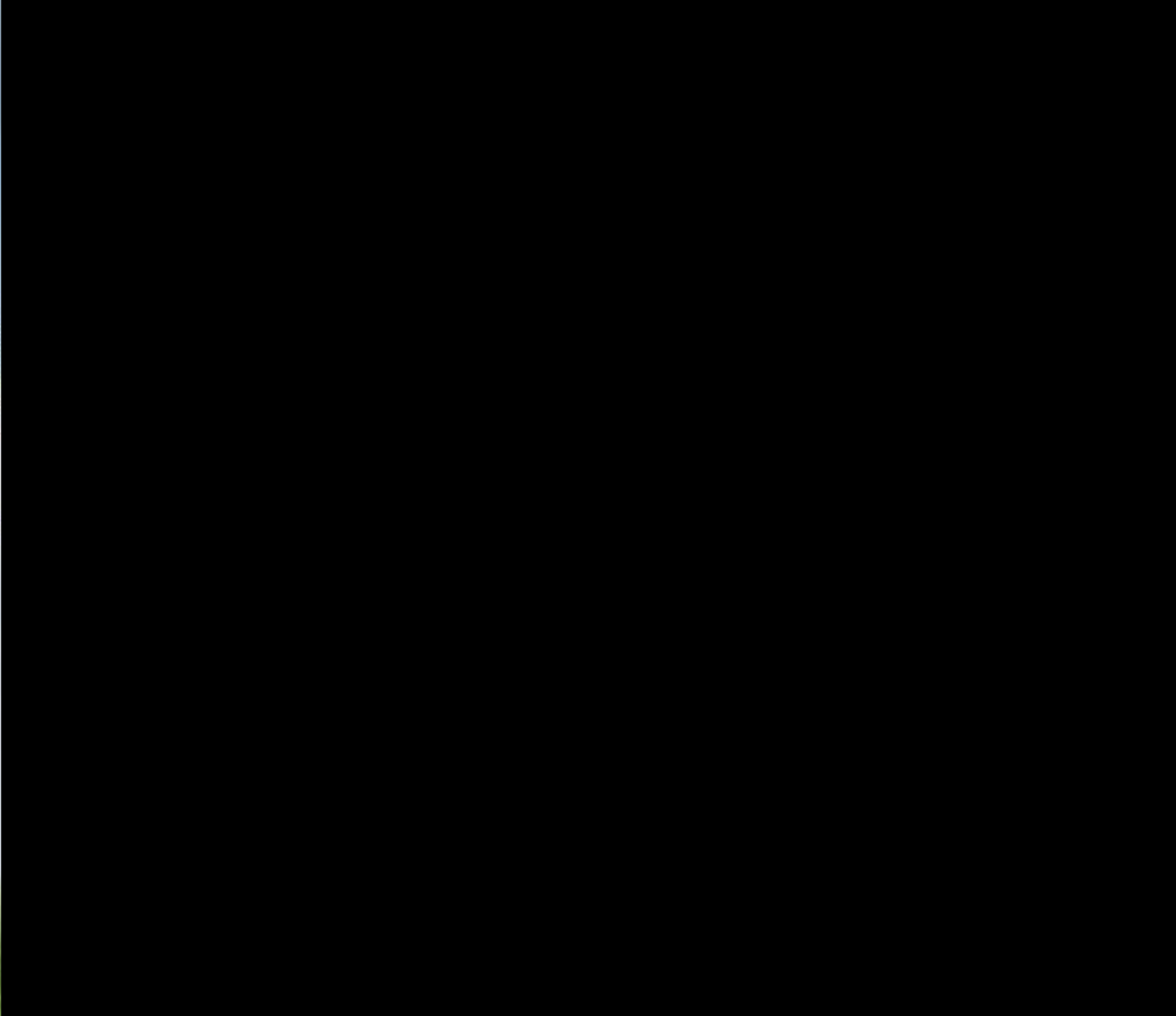
Decoding the diversity of neuronal nanostructures with AI-assisted nanoscopy

Flavie Lavoie-Cardinal

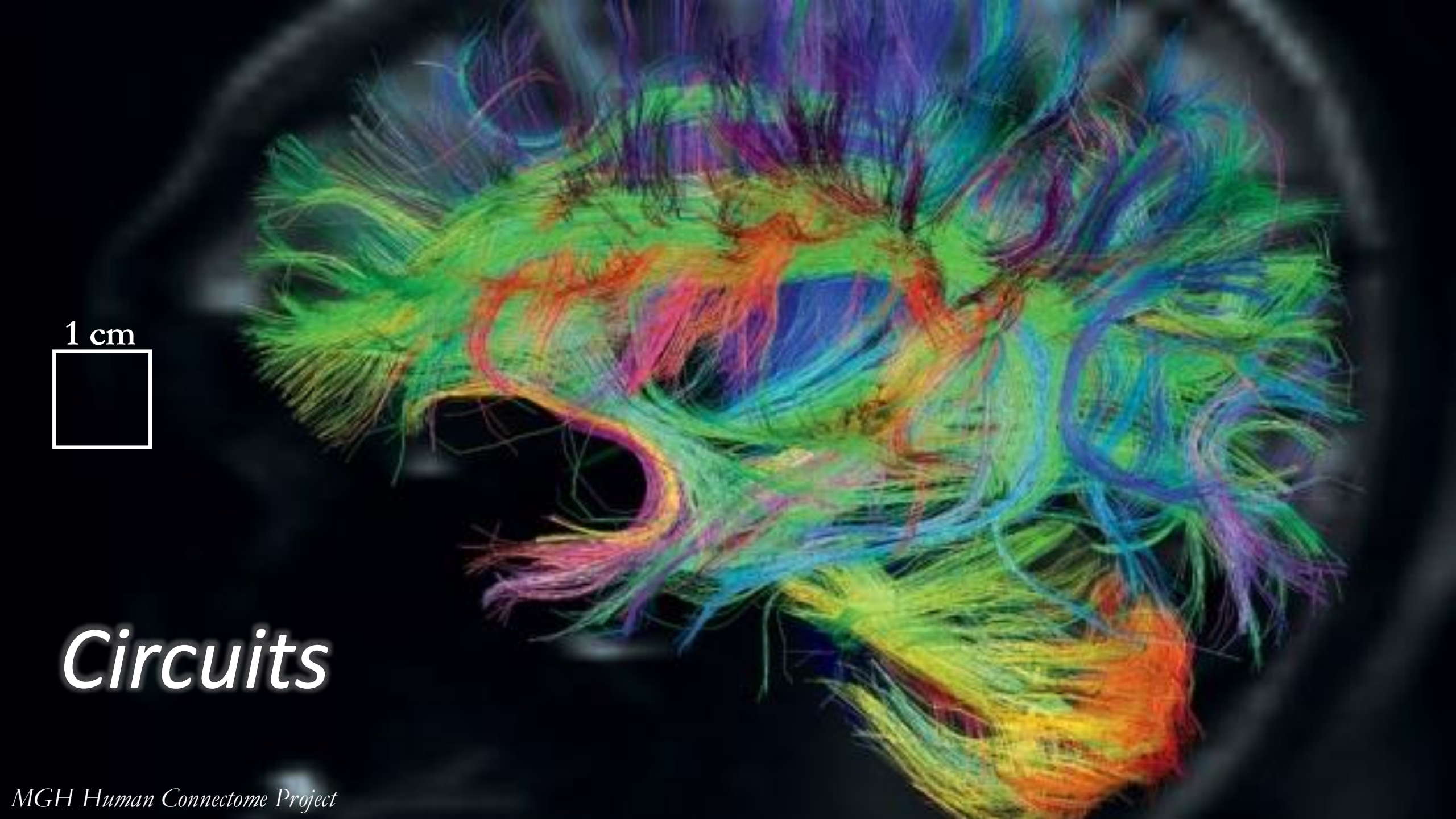
Canada Research Chair in Intelligent Nanoscopy of Cellular Plasticity

Université Laval, Québec, Canada

CERVO Brain Research Center, Institute for Intelligence and Data

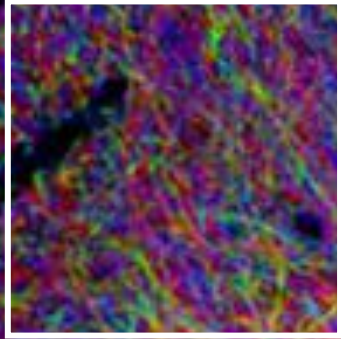


1 cm

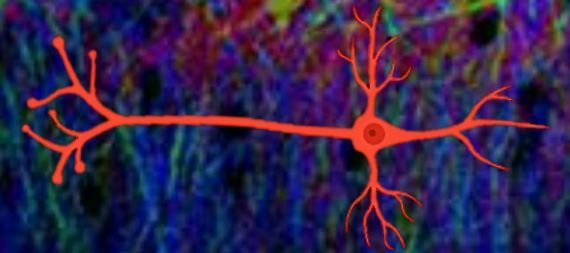


Circuits

100 μm



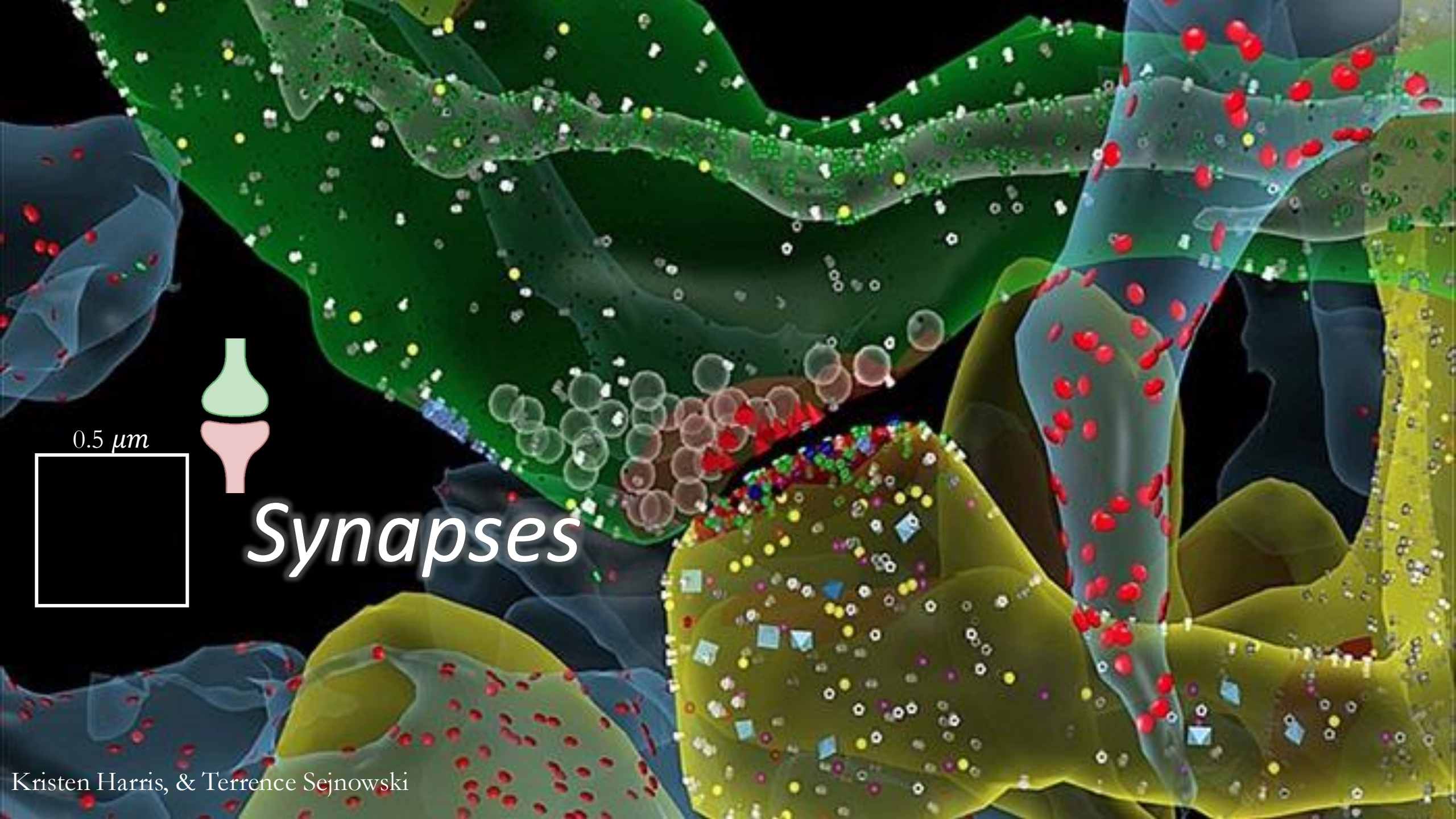
Neurons



J. Lichtman, *Brainbow*

10 μm

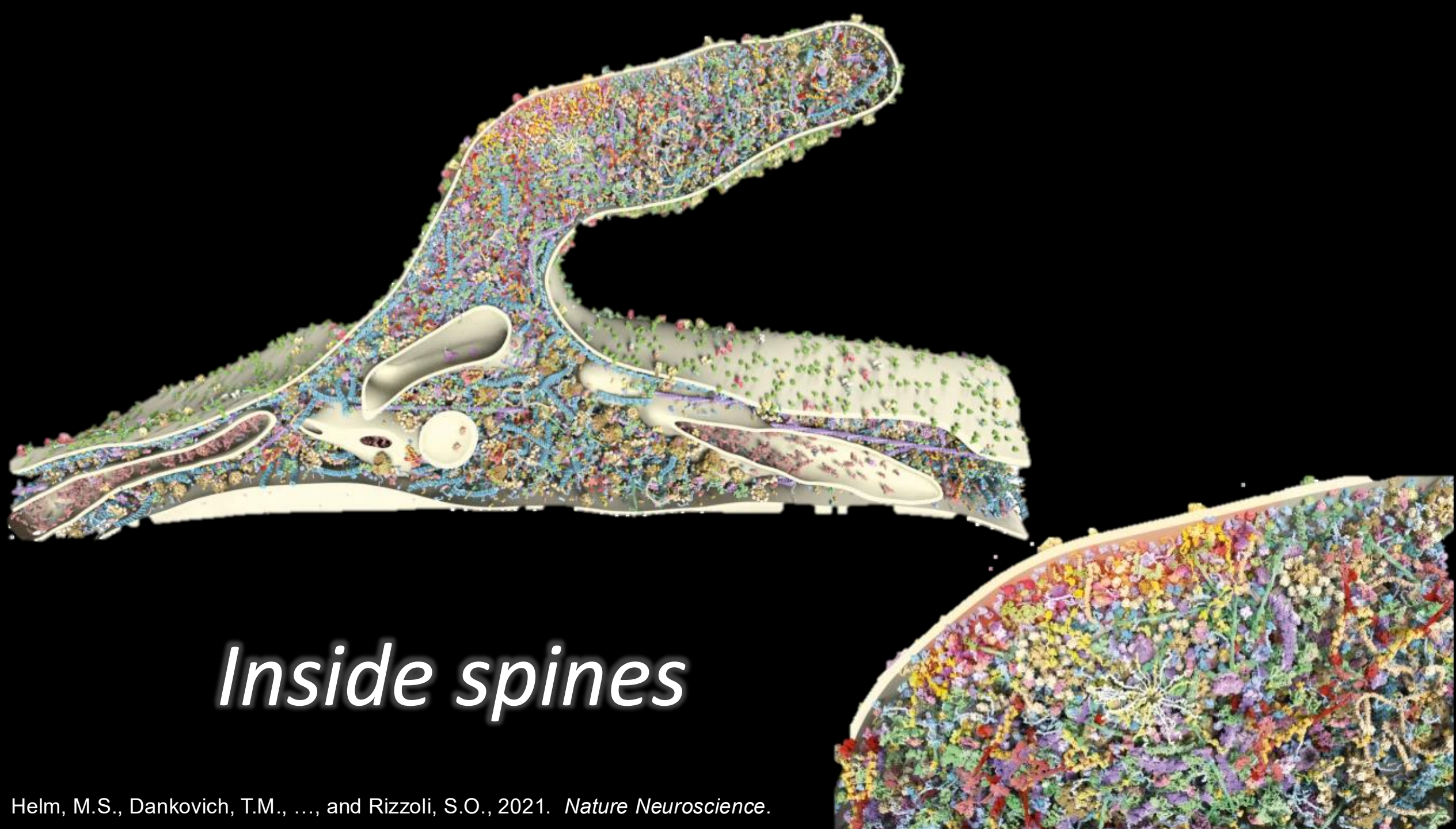
Connectivity



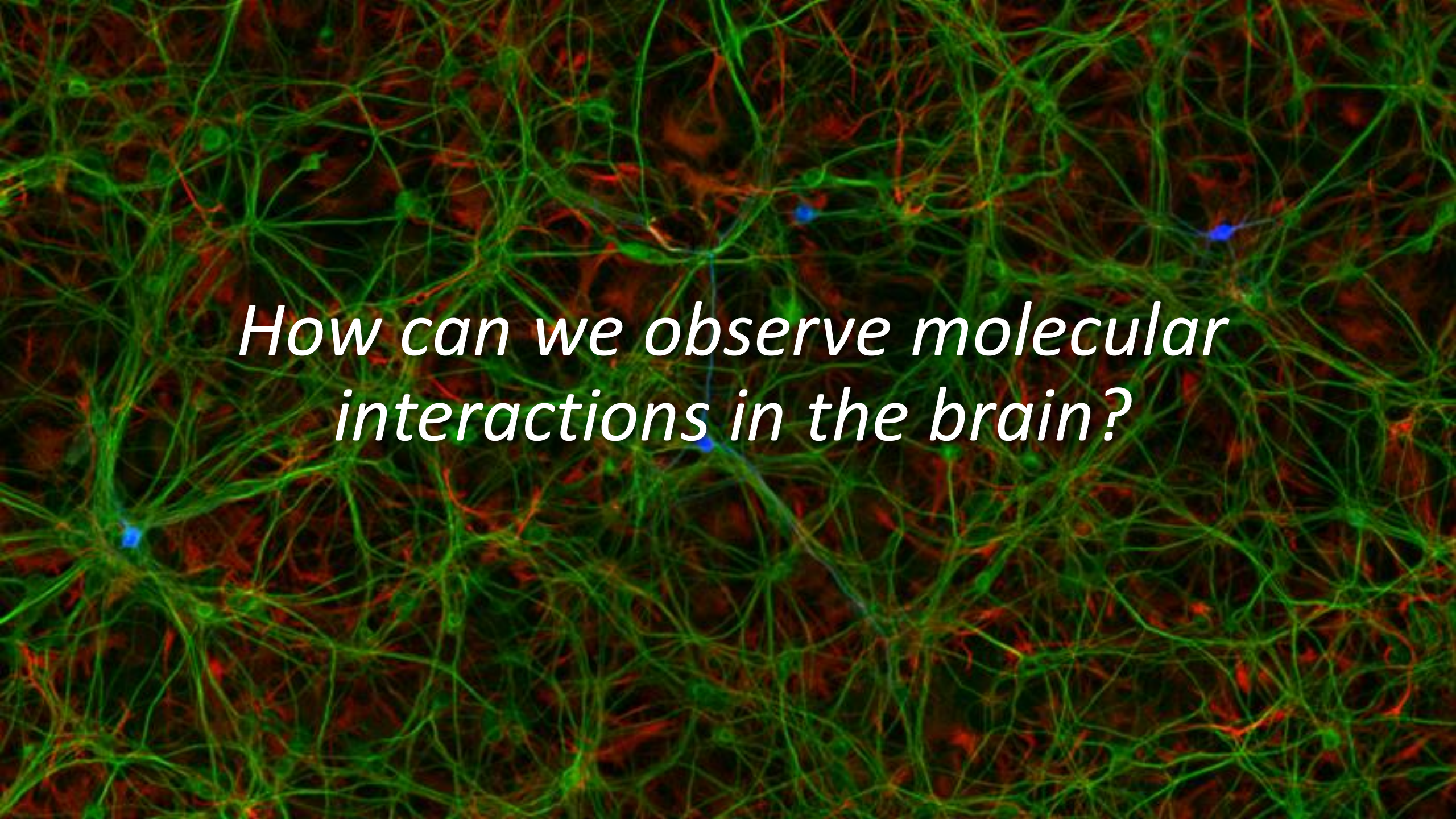
0.5 μm



Synapses

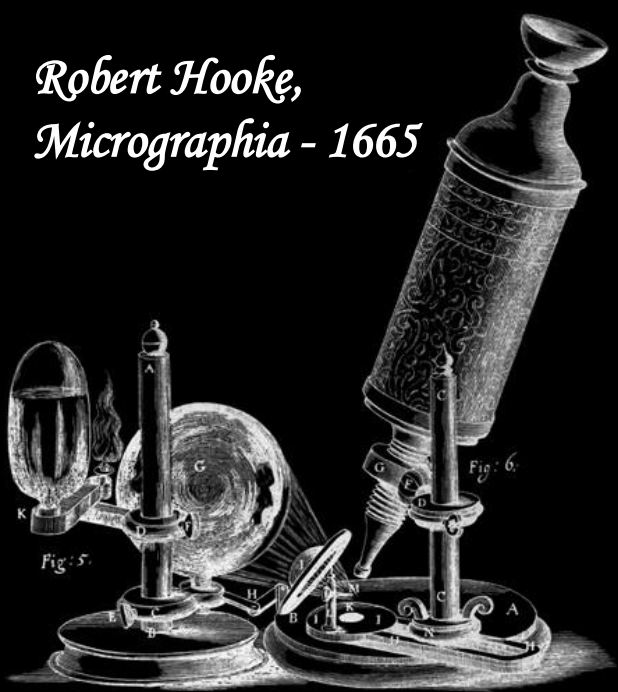


Inside spines

A fluorescence micrograph showing a dense, interconnected network of neural fibers. The fibers are primarily green, with some red fibers interspersed. Several small, bright blue spots are visible, likely representing cell nuclei or specific molecular markers. The background is dark, making the colored fibers stand out.

*How can we observe molecular
interactions in the brain?*

*Robert Hooke,
Micrographia - 1665*



(a)



(b)

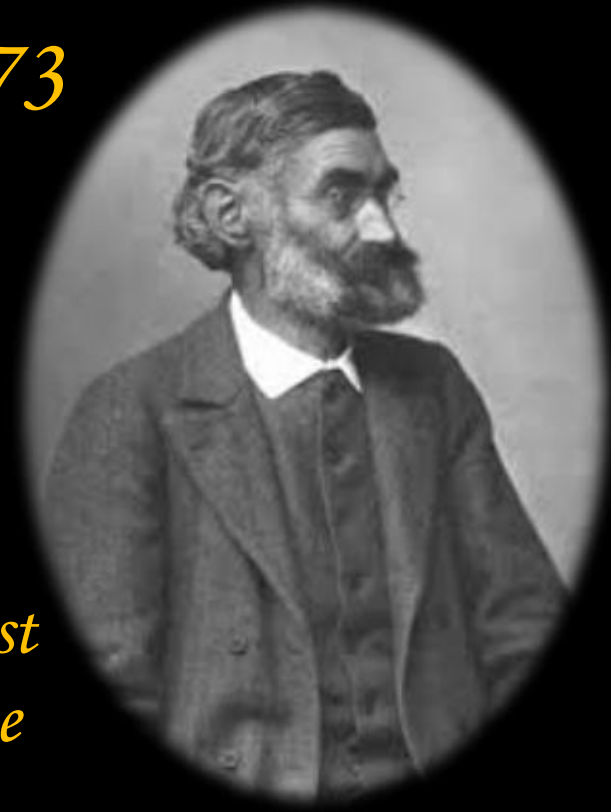


*Carl Zeiss & Ernst Abbe
Around 1870*



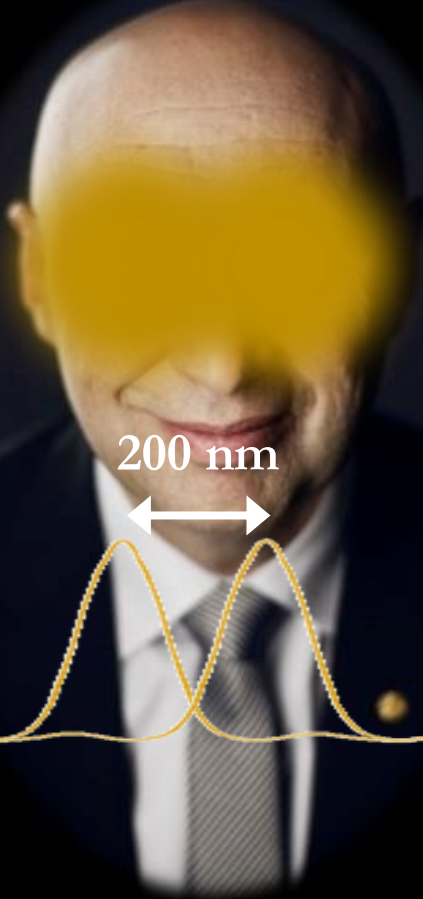
1873

Ernst
Abbe



$$d = \frac{\lambda}{2n \sin \alpha}$$

1994



Stefan W. Hell

200 nm

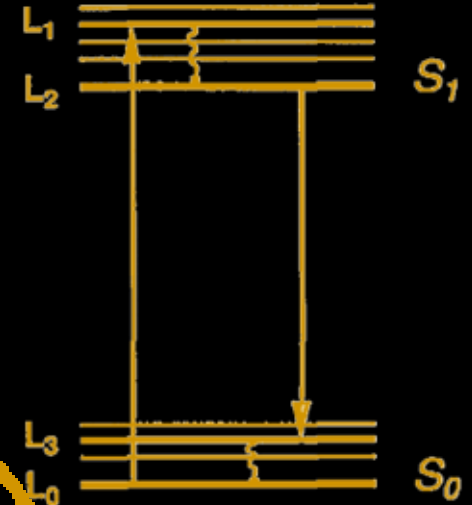
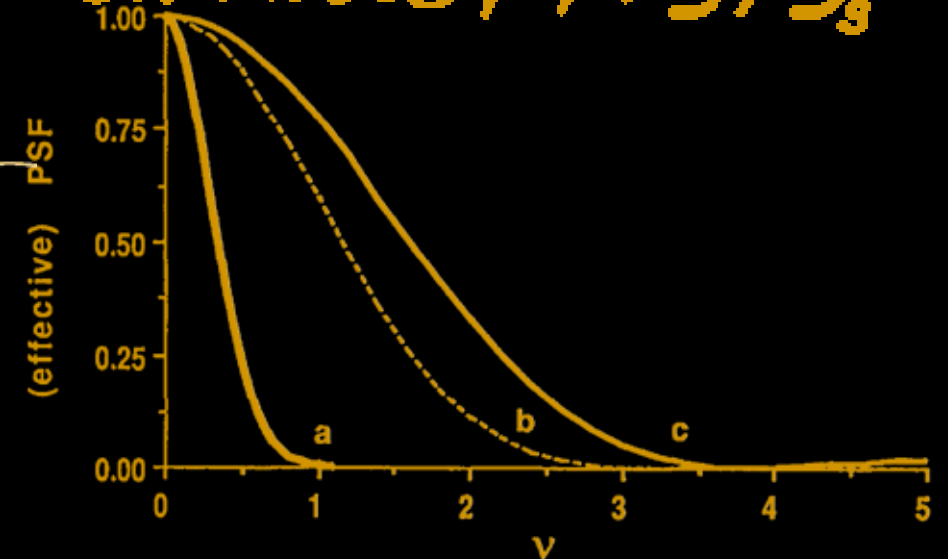
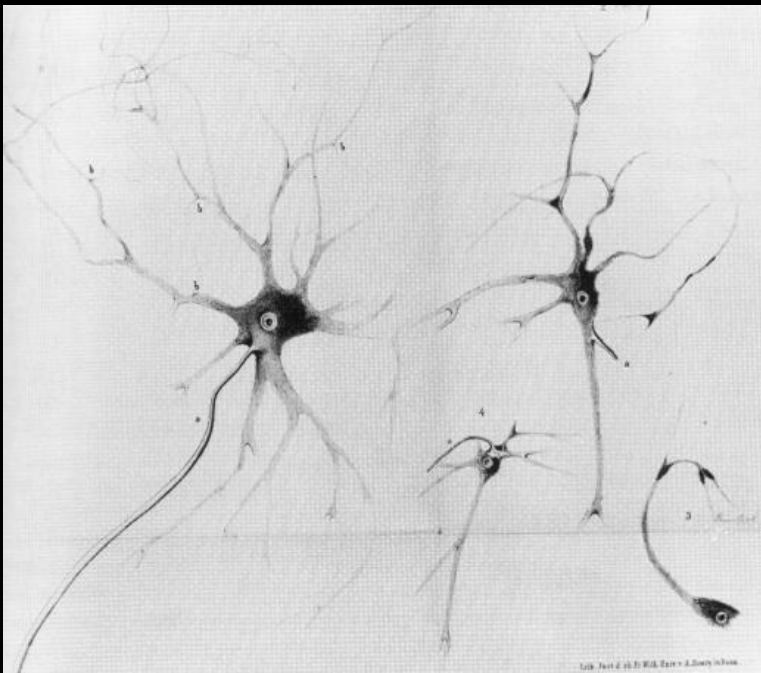


Fig. 1. Energy levels of a typical fluorophore.

© 1994 Optical Society of America

$$d = \frac{\lambda}{2n \sin \alpha \sqrt{1 + J/J_s}}$$





Neurons

Otto F.K. Deiters - 1865

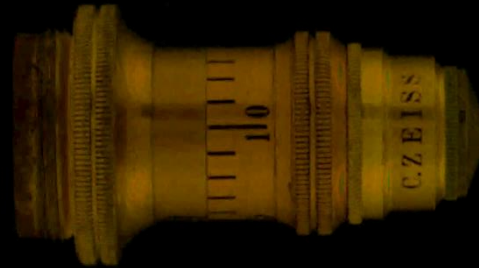
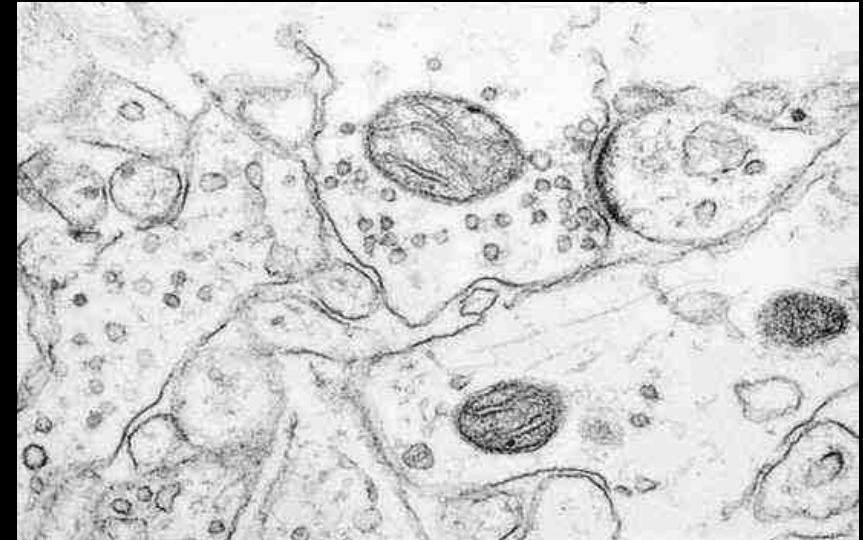
Spines and Dendrites

Ramón y Cajal - Around 1900

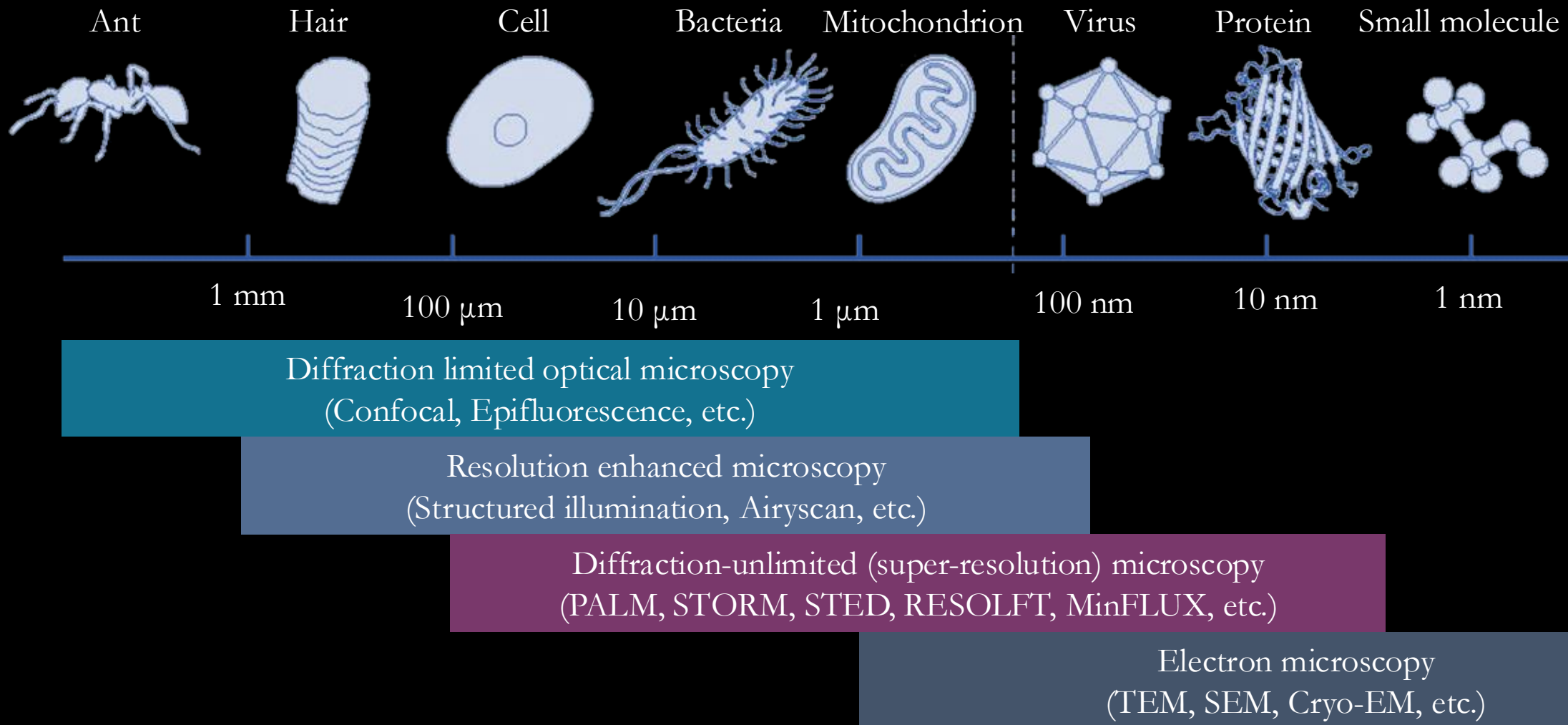


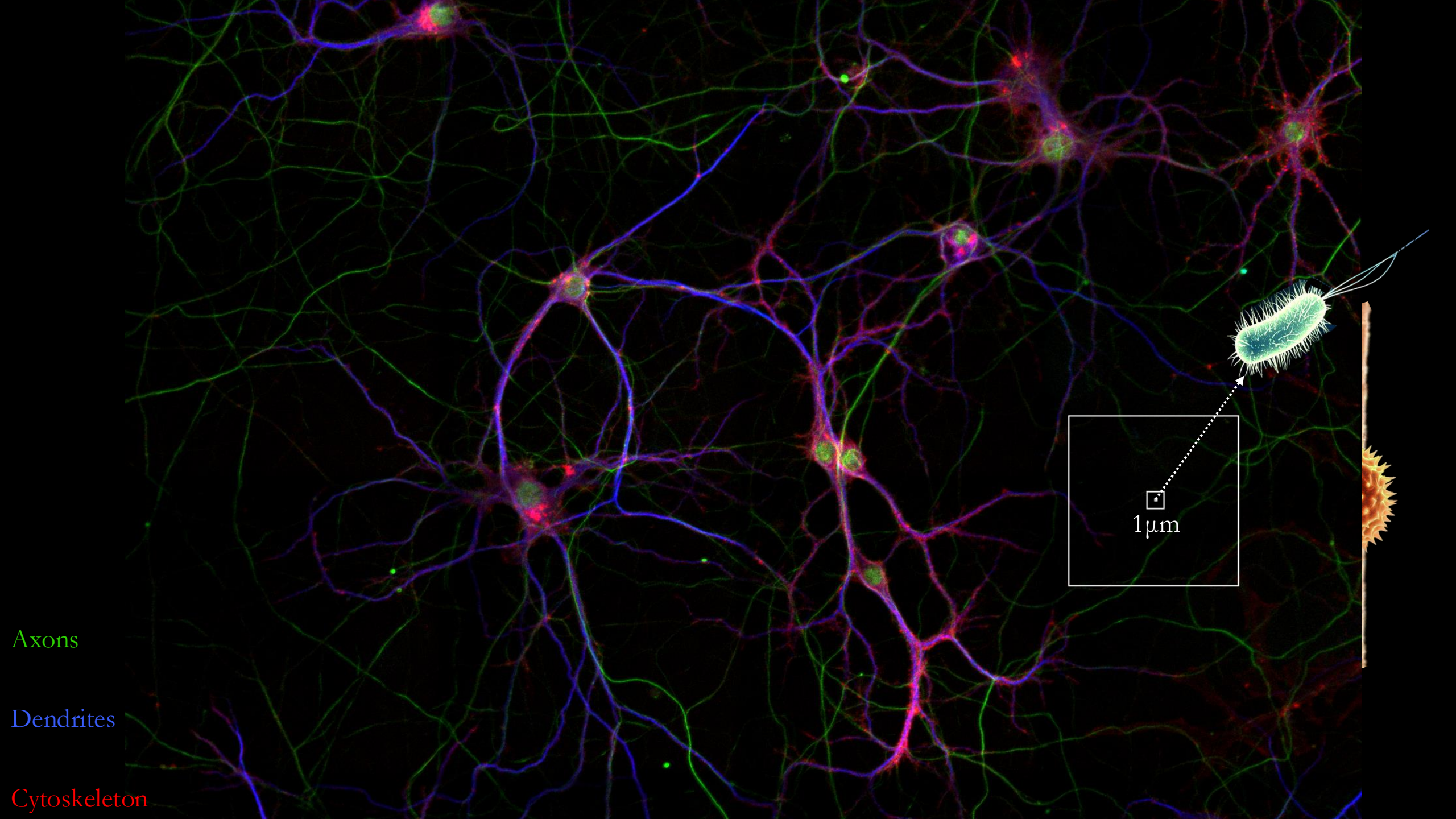
Synapses

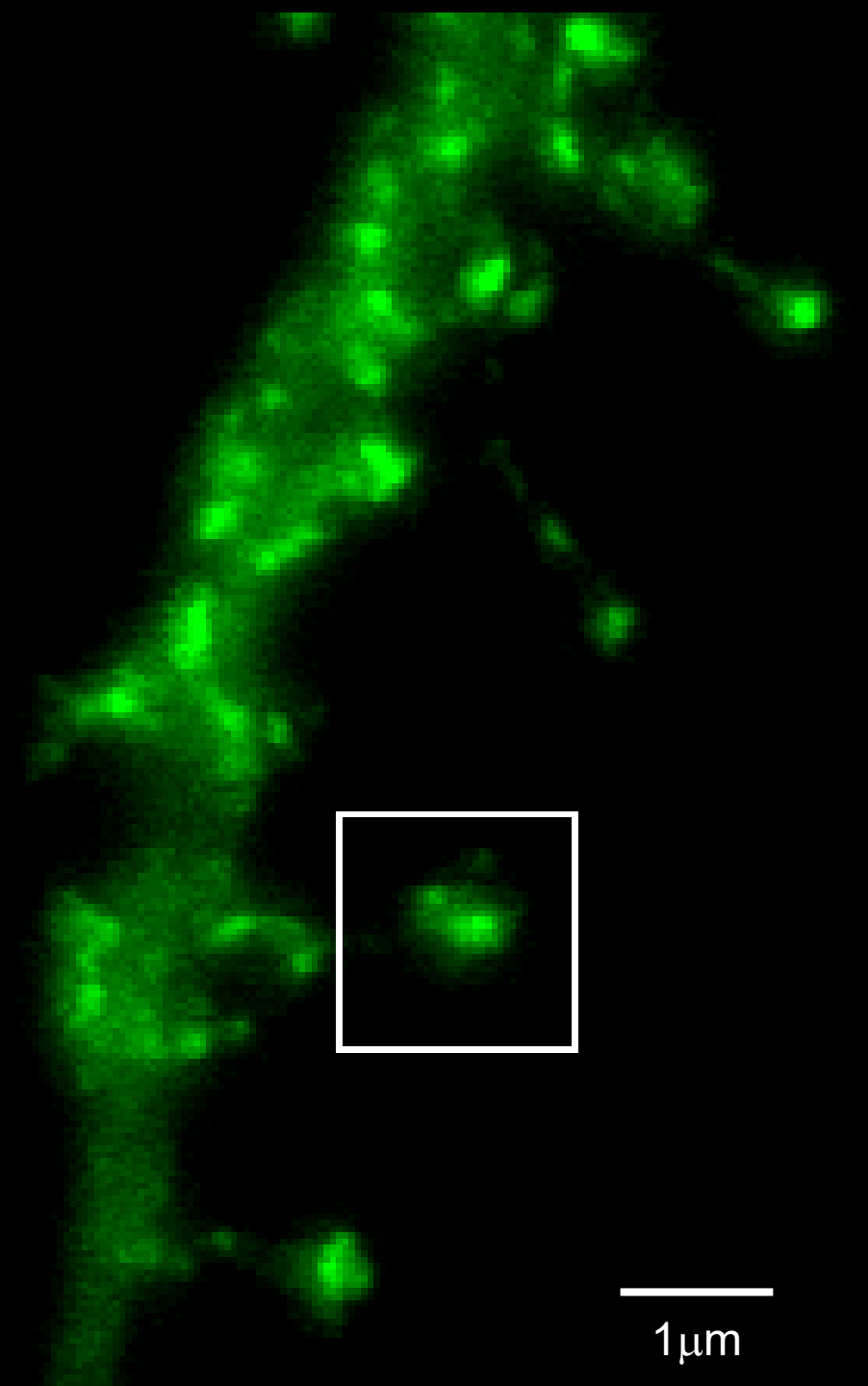
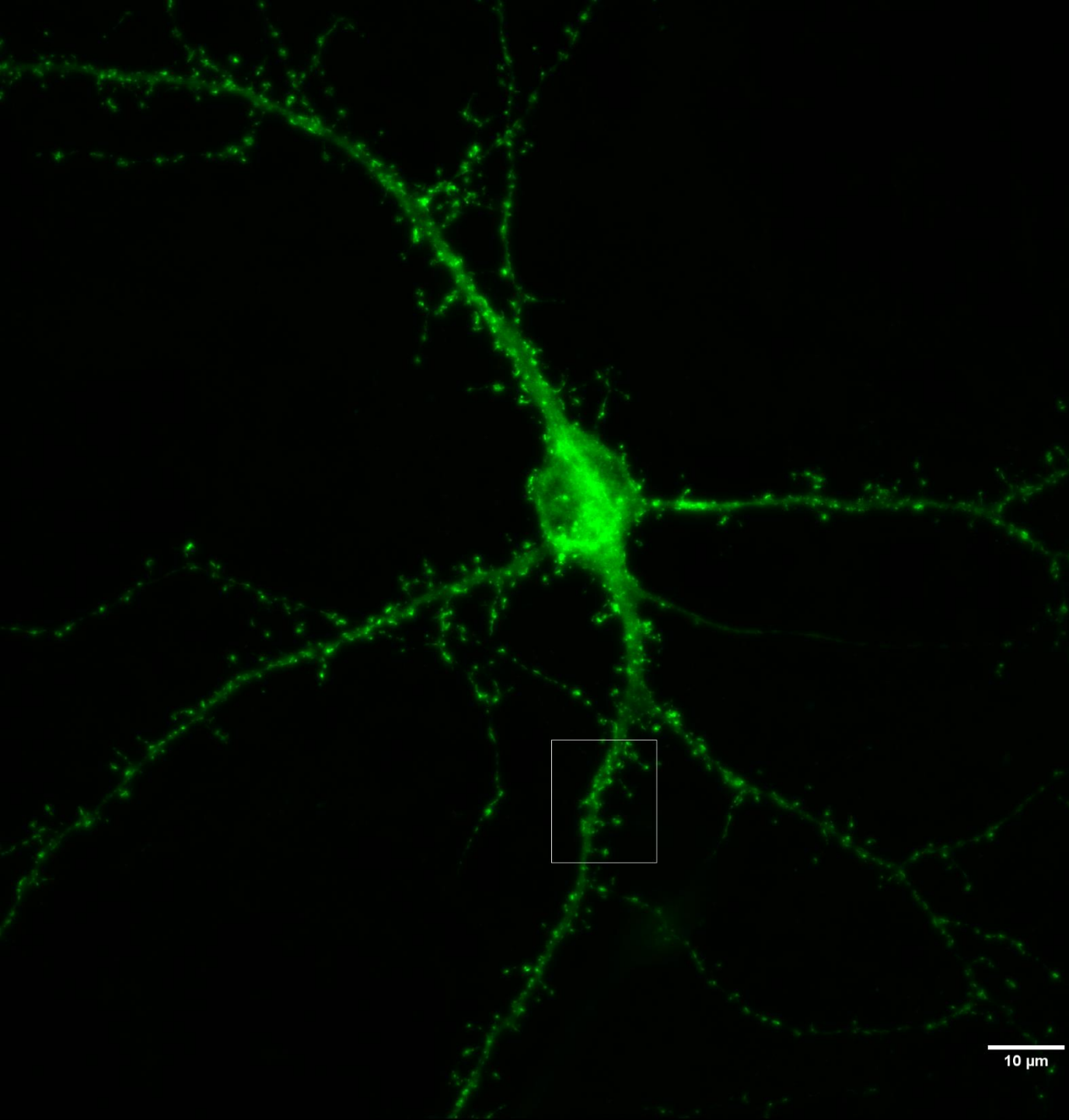
*George Palade, Eduardo de Robertis
and George Bennett - 1954*

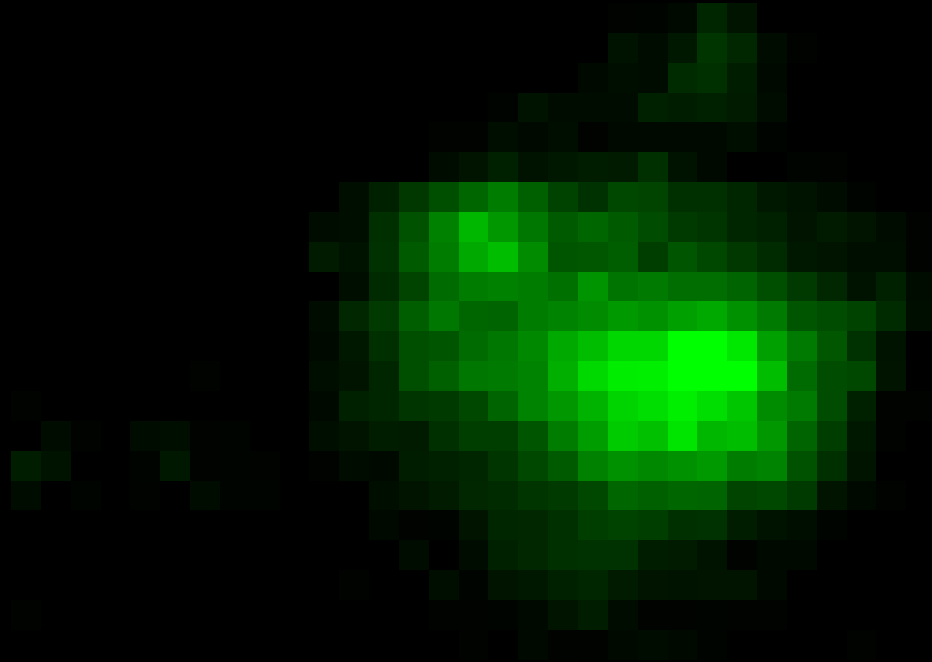


Optical microscopy

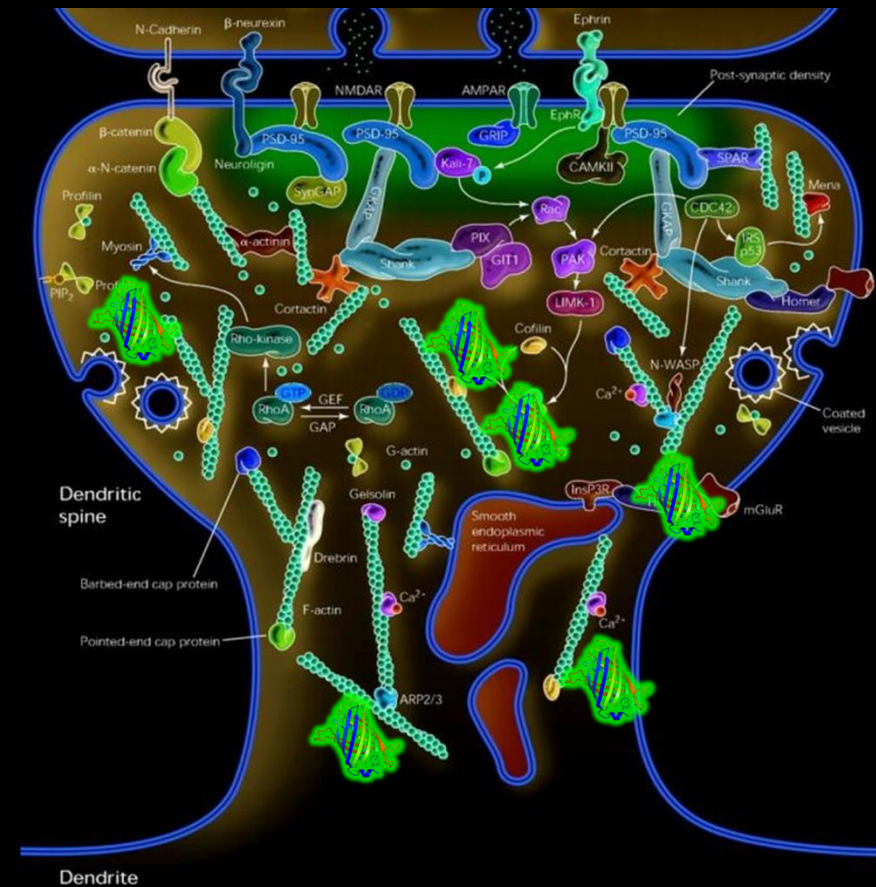








Spine

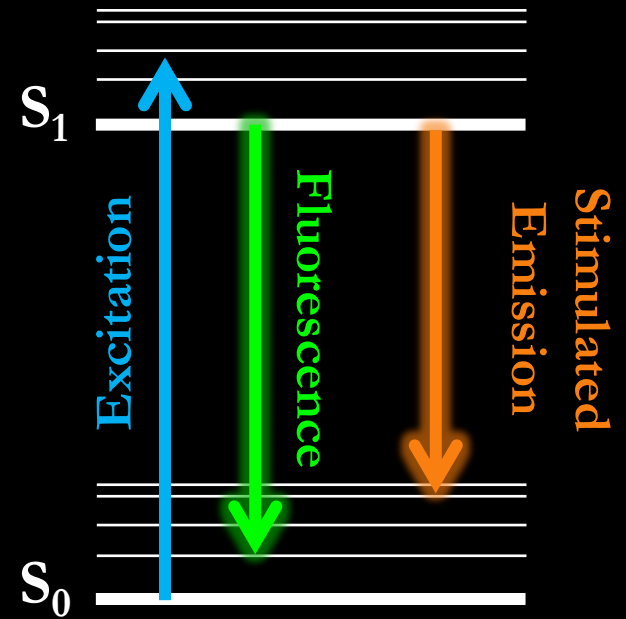


1μm

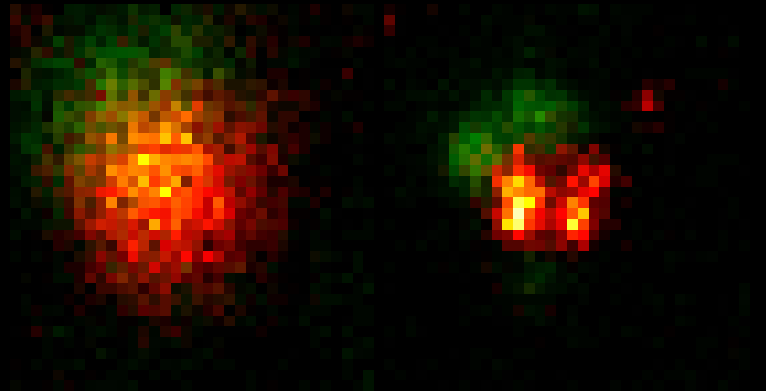
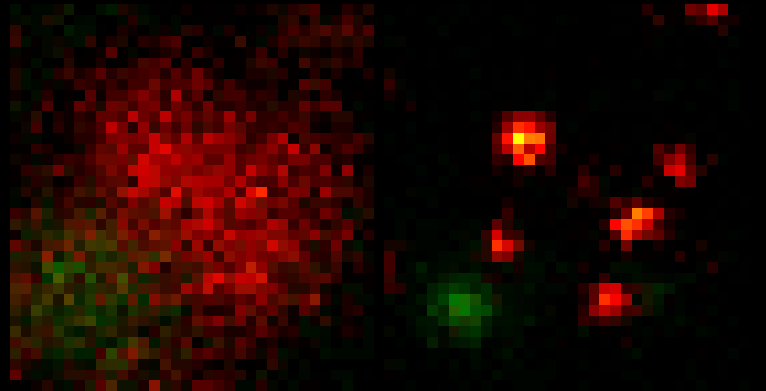
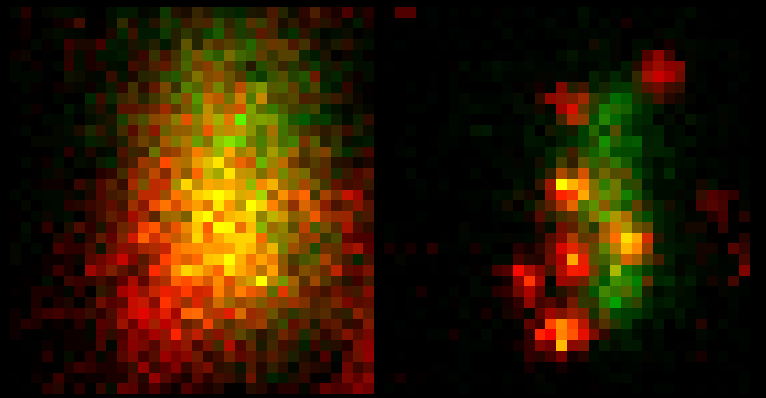
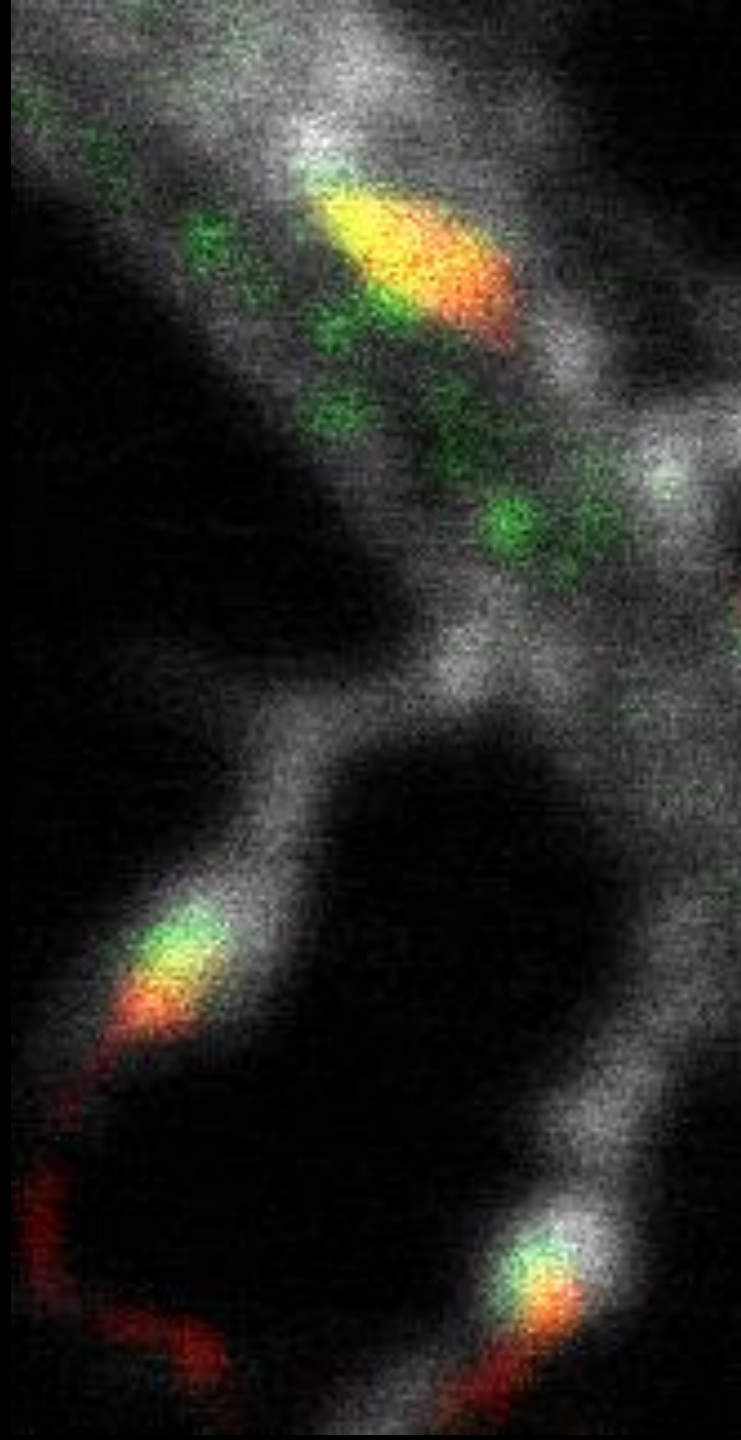
STimulated Emission Depletion Microscopy (STED)

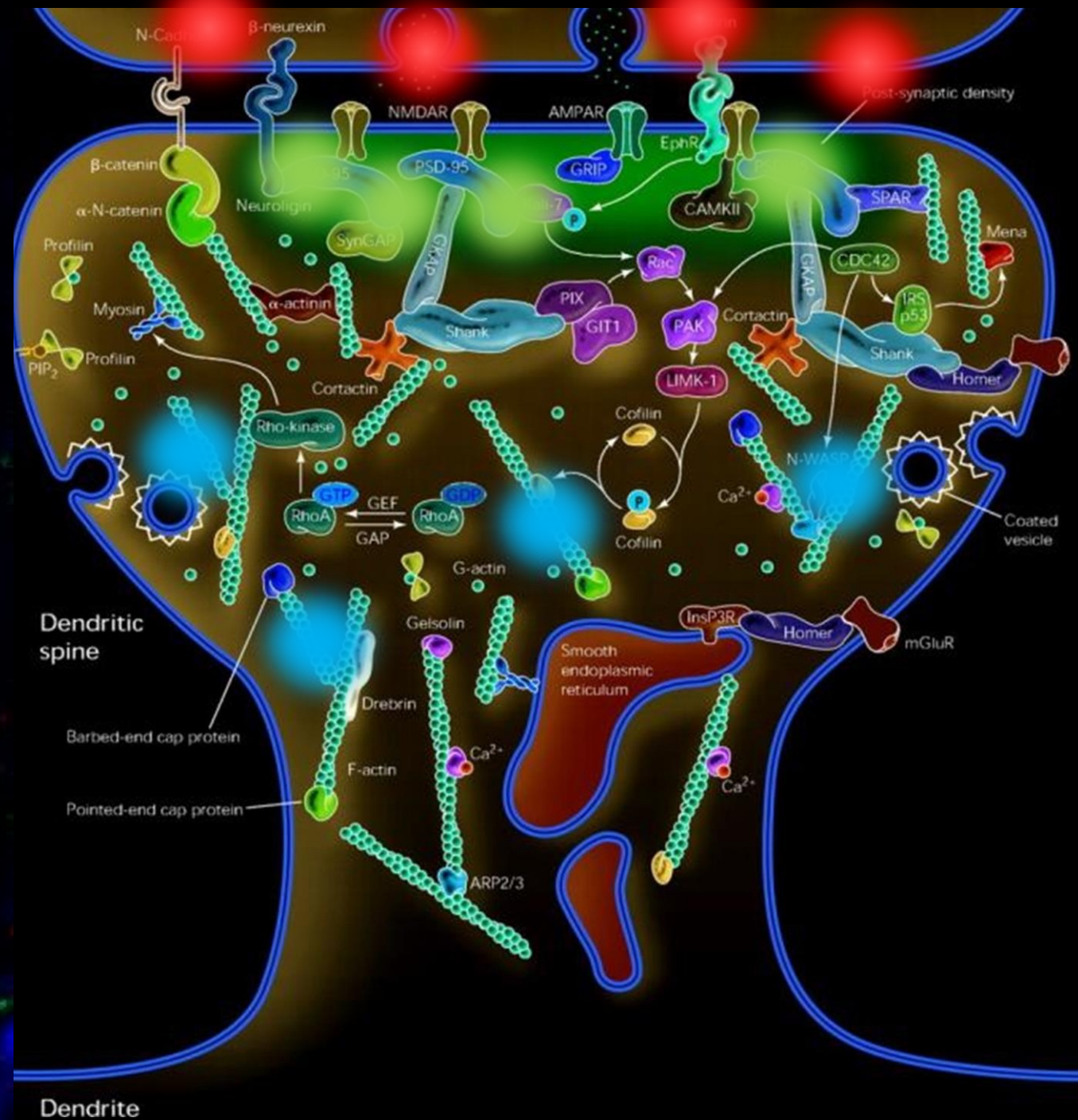
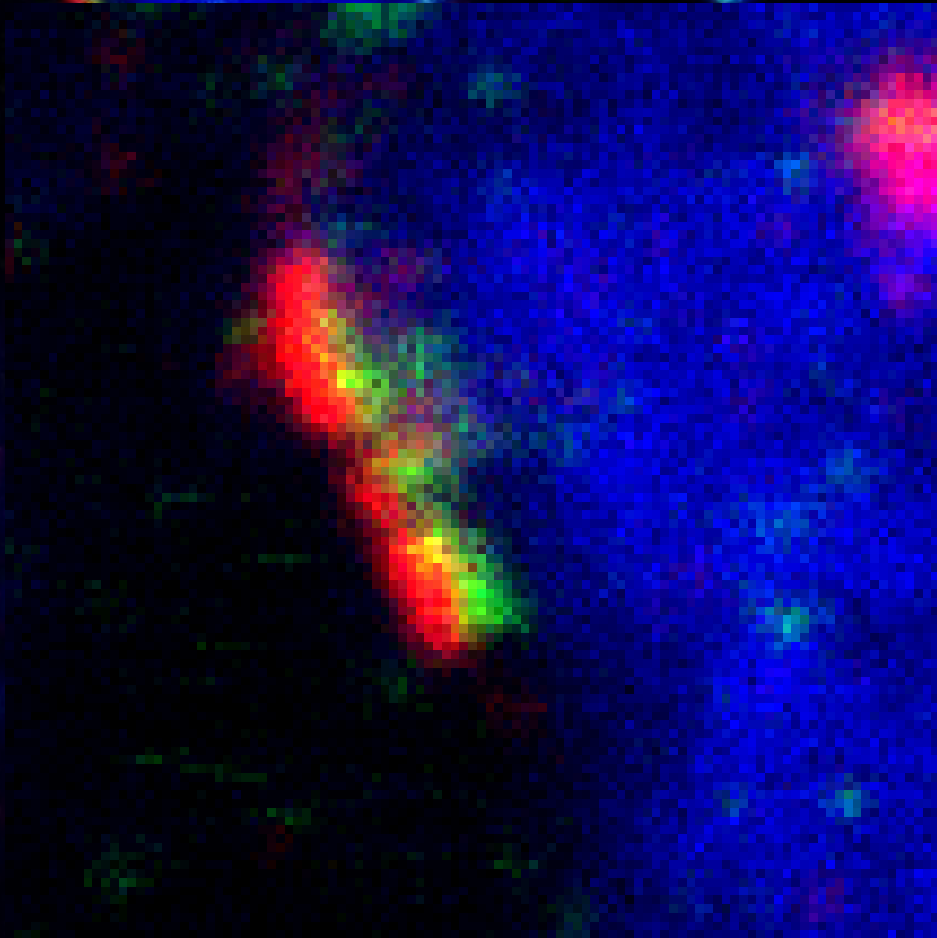
Confocal microscopy

STED microscopy

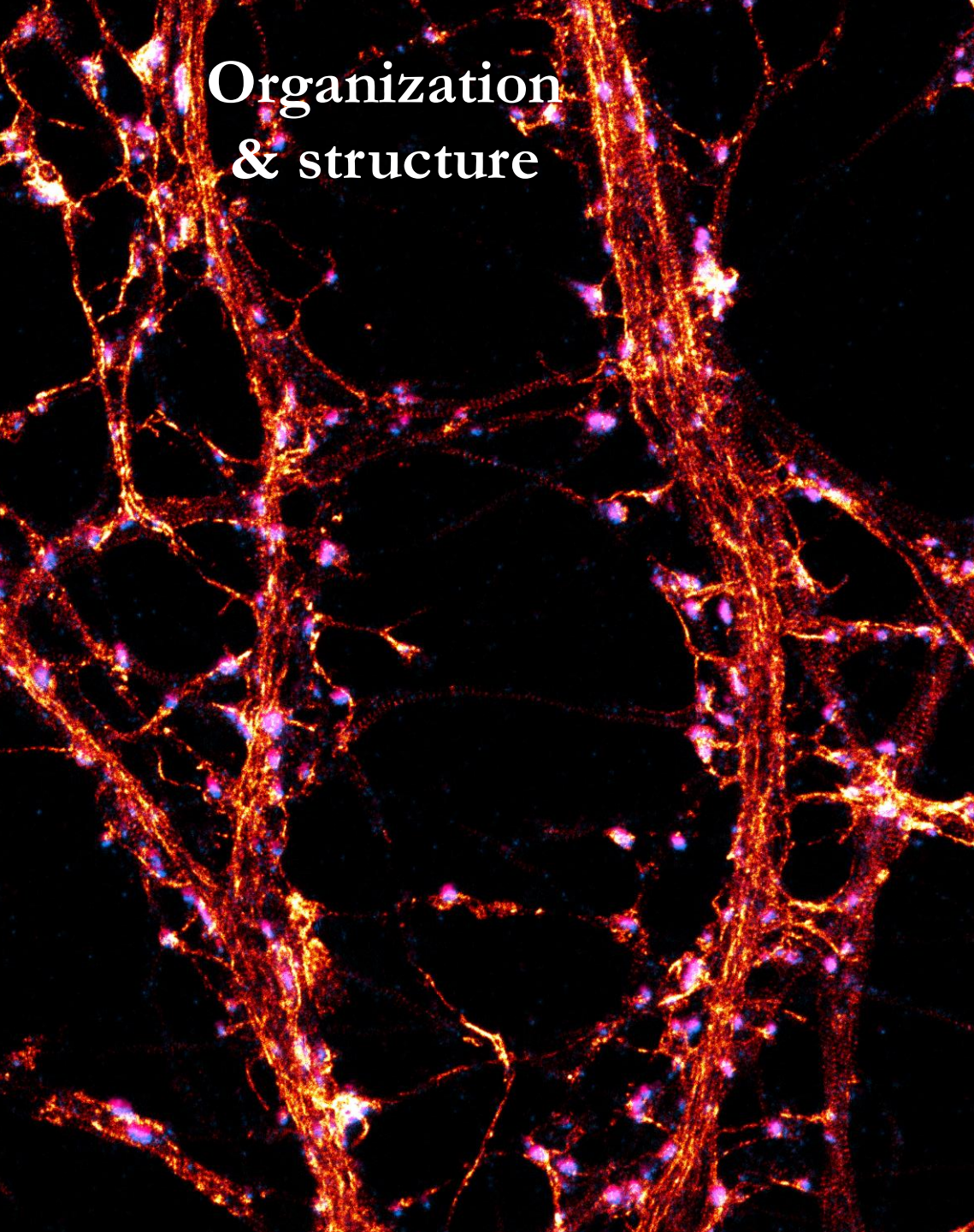


STED imaging of neuronal proteins

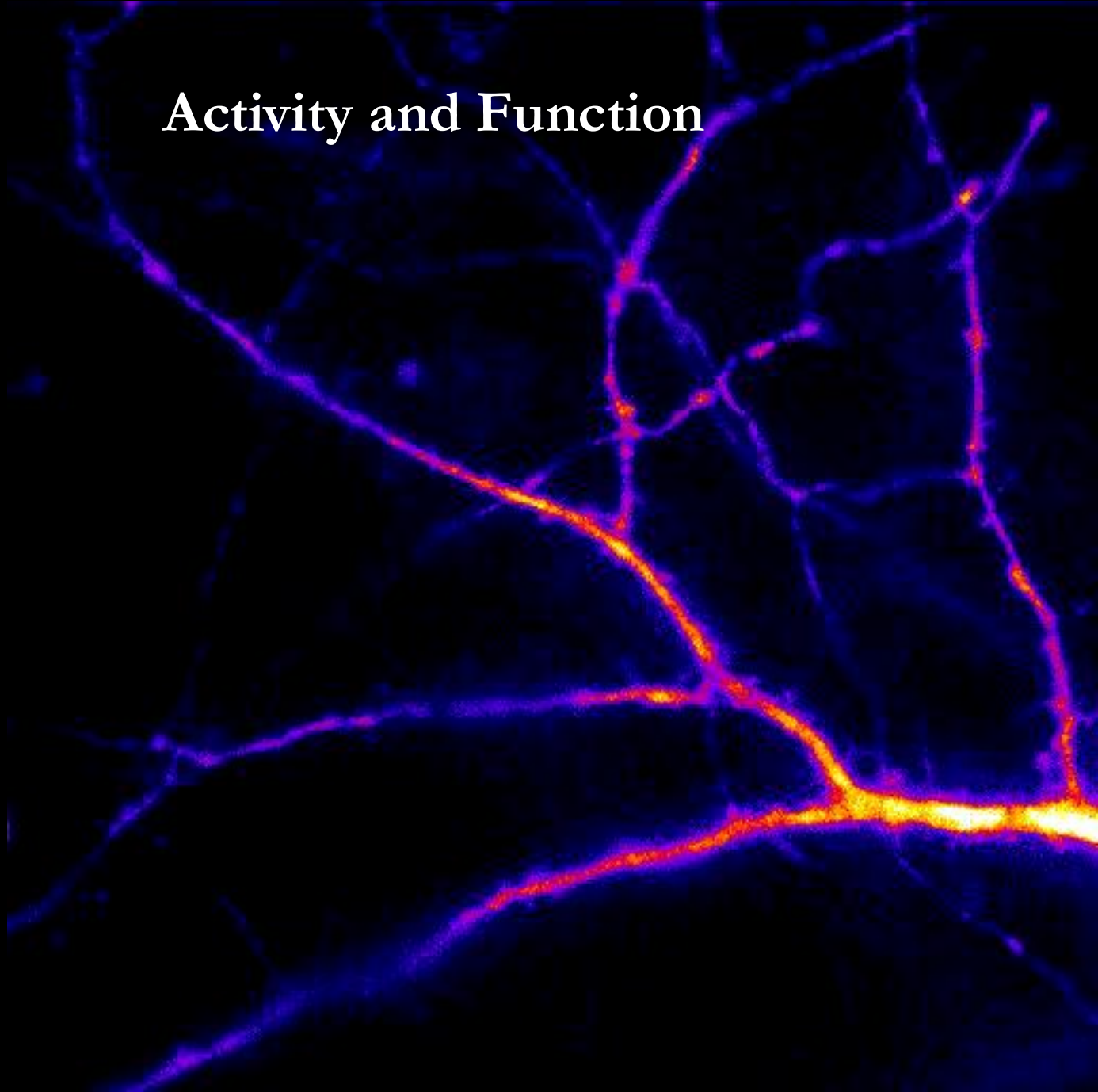


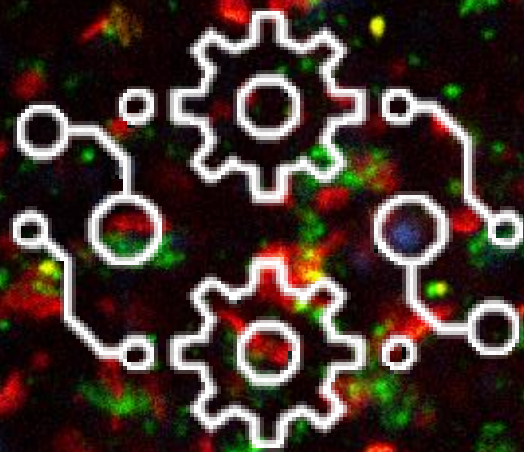


Organization
& structure



Activity and Function

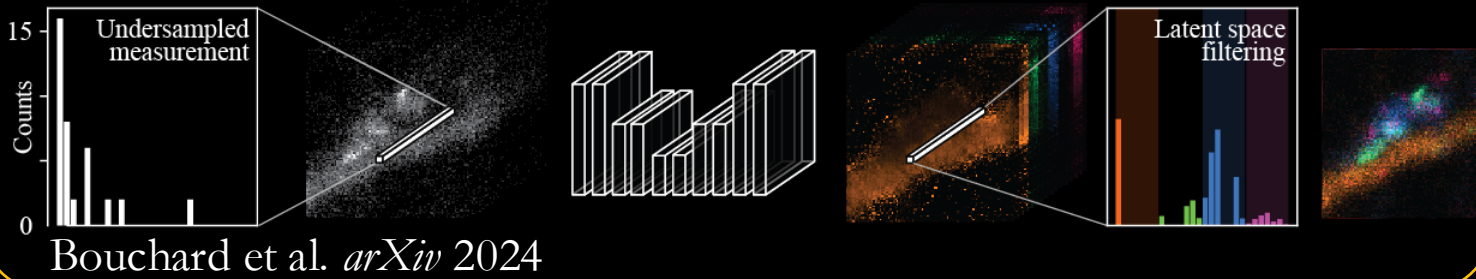




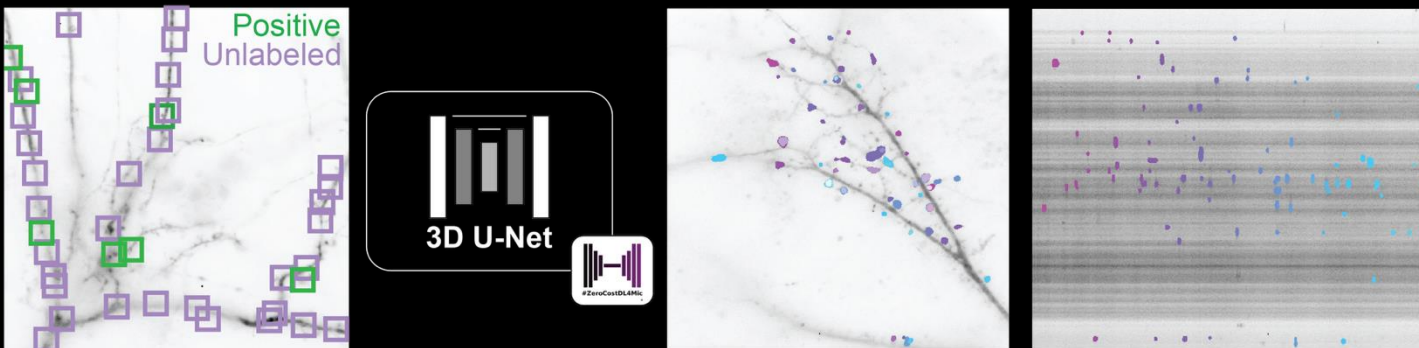
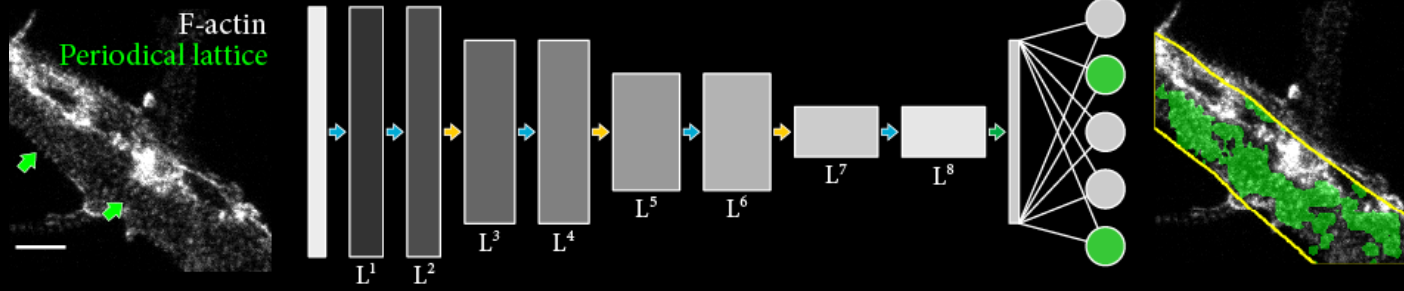
AI-Assisted Microscopy

Quantitative analysis strategies in bioimaging

Signal Unmixing in the low photon regime

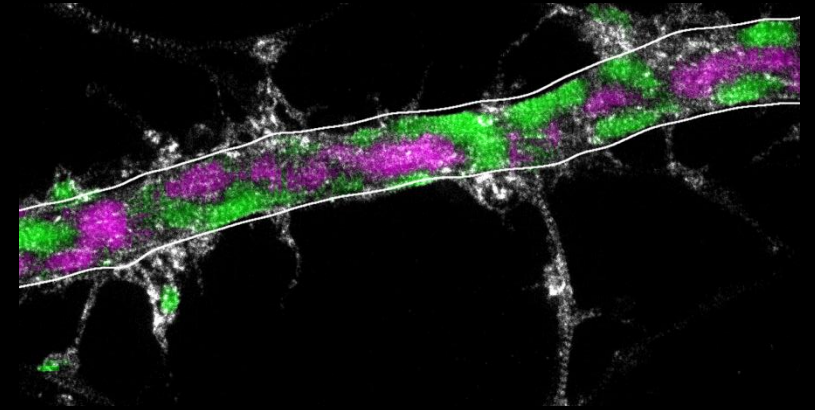


Reducing the annotation burden

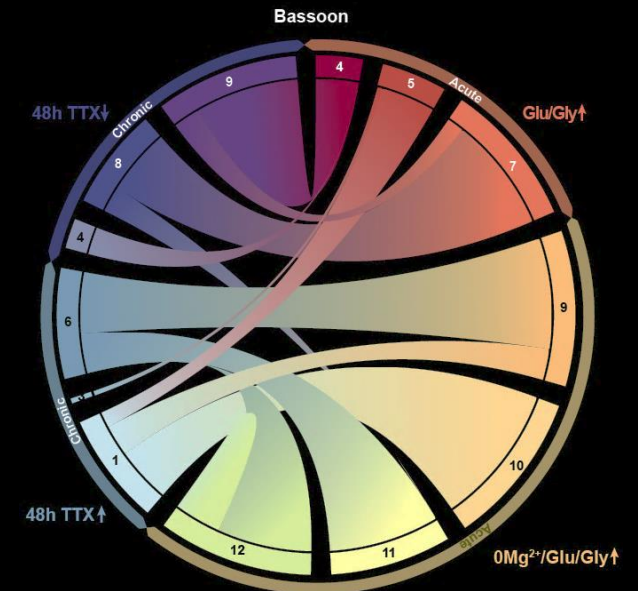


Beaupré, Bilodeau et al. *Digital Discovery* 2024

Decoding diversity in bioimaging datasets



Lavoie-Cardinal et al. *Scientific Reports* 2020

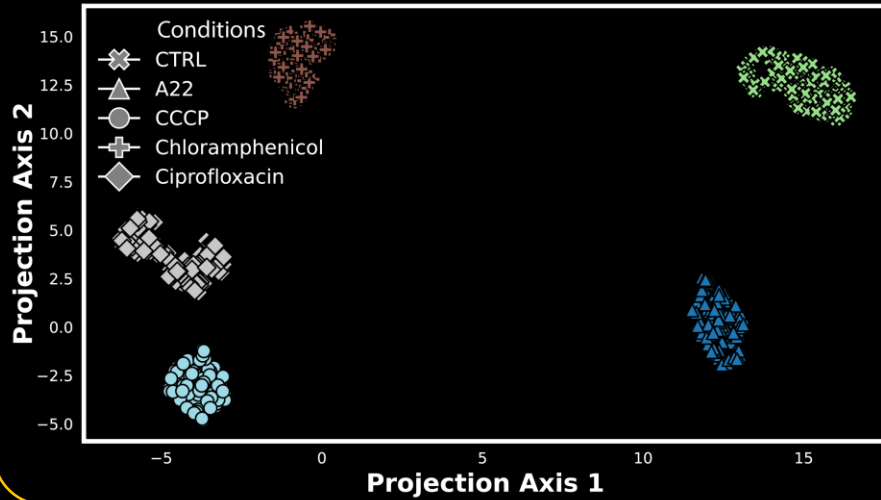


Wiesner et al. *Frontiers in Neural Circuits* 2022

Quantitative analysis strategies in bioimaging

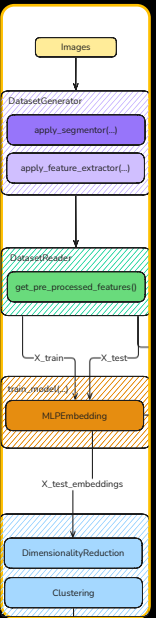
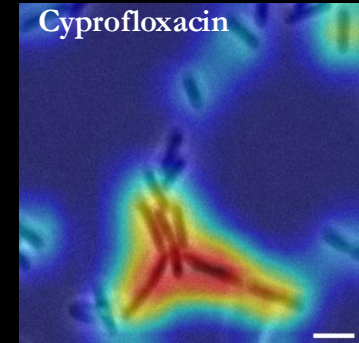
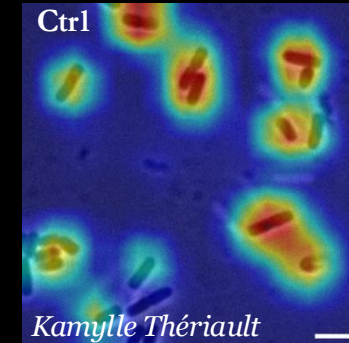


Mode of action discovery



	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
CTRL	1	271	0	0	0
A22	219	0	0	1	0
CCCP	0	0	0	23	197
Chloramphenicol	0	0	220	0	0
Ciprofloxacin	0	0	0	220	0

Robust and interpretable pipelines for high-throughput microscopy



Generalizable models for bioimaging

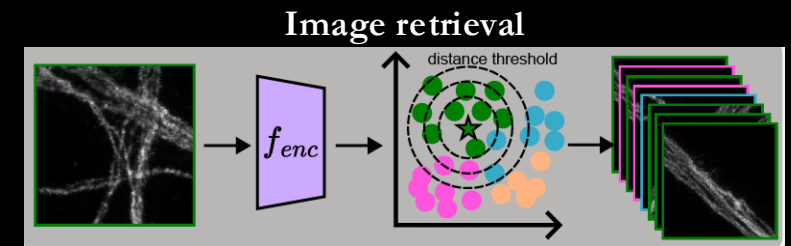
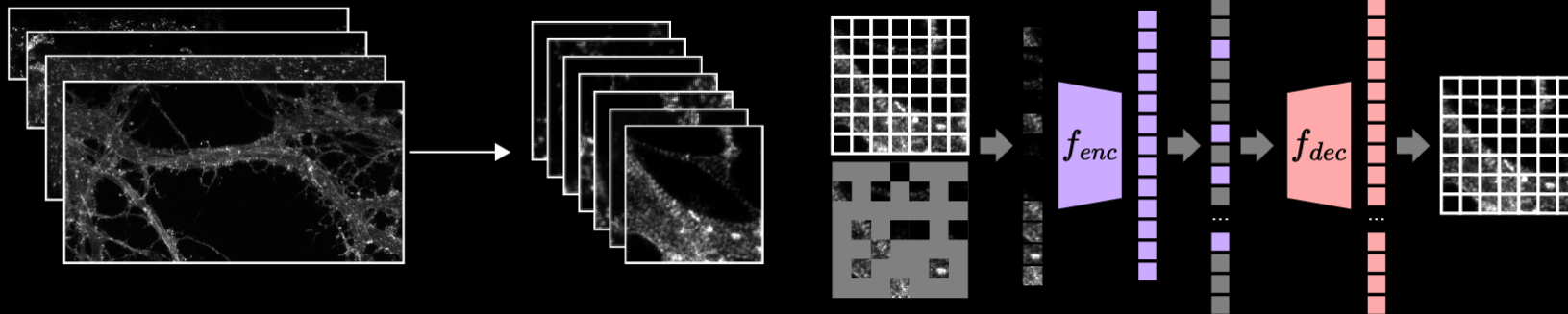
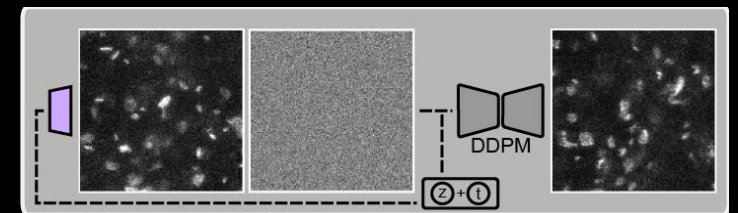
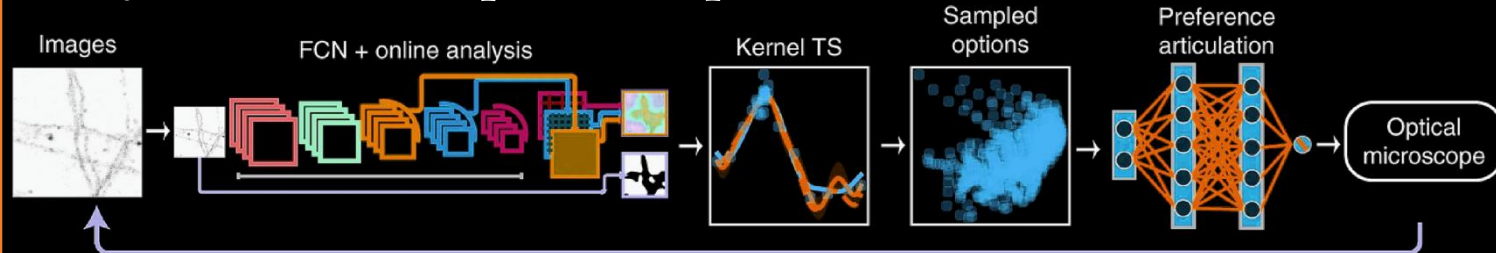


Image generation & Pattern discovery



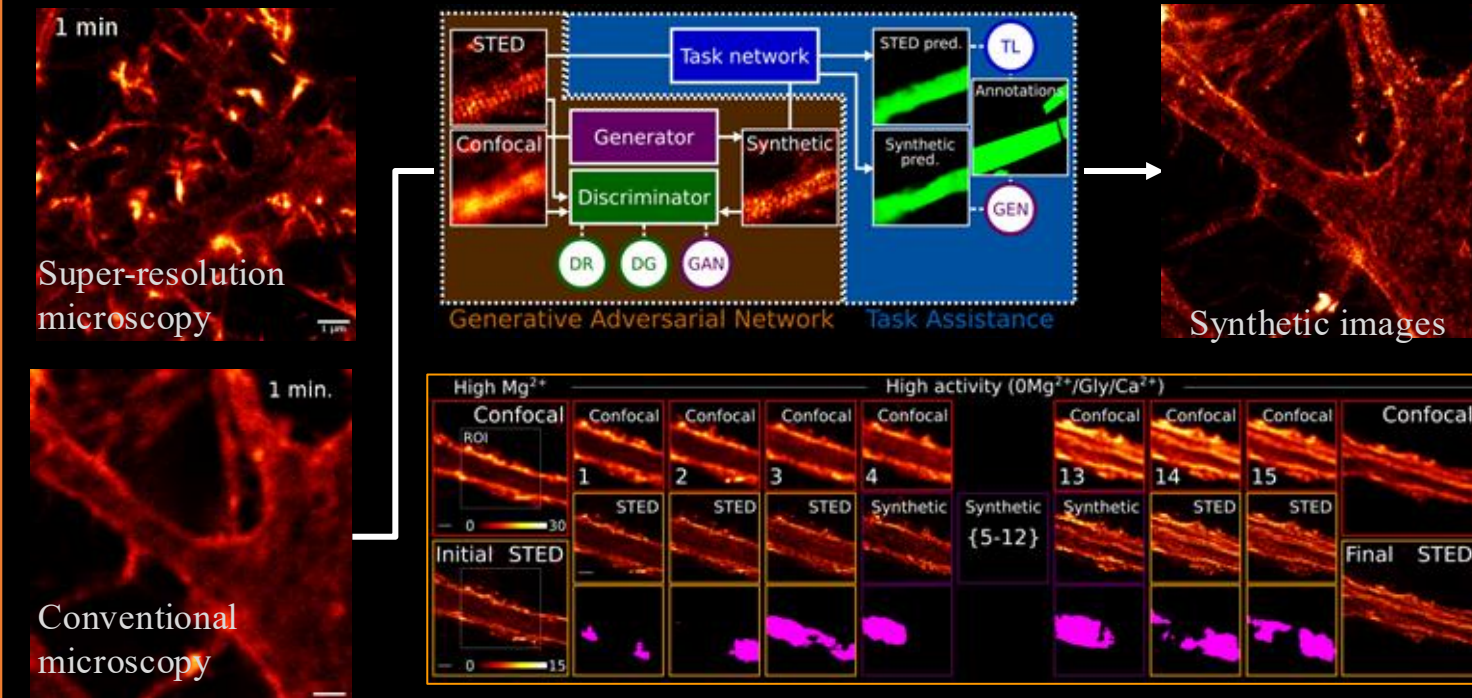
Bioimaging acquisition guidance

Fully automated acquisition optimization with ML



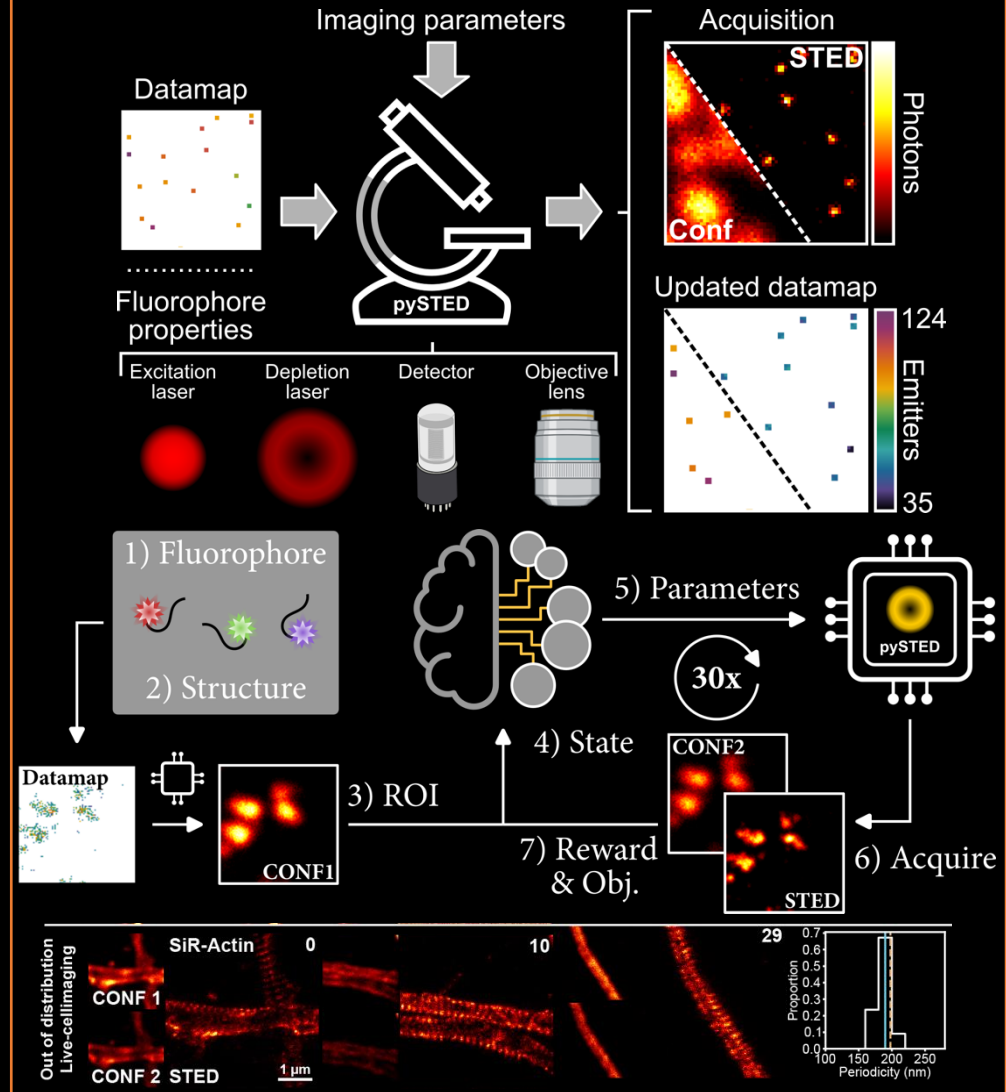
Durand et al. *Nature Communications* 2018

Acquisition guidance with generative AI



Bouchard et al. *Nature Machine Intelligence* 2023

Training RL agents to perform bioimaging task



Bilodeau et al. *Nature Machine Intelligence* 2024

Synaptic remodeling at the nanoscale



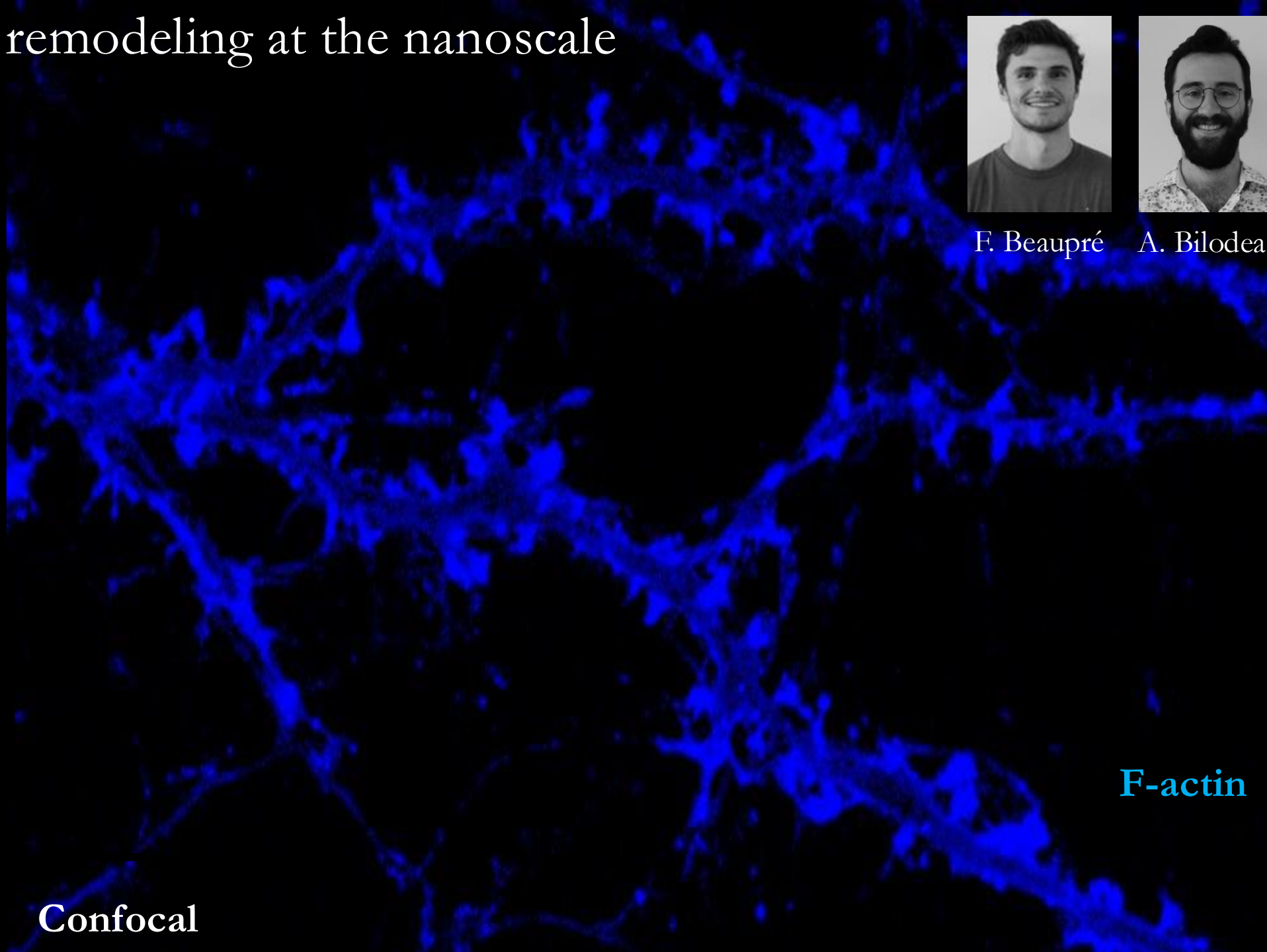
F. Beaupré



A. Bilodeau



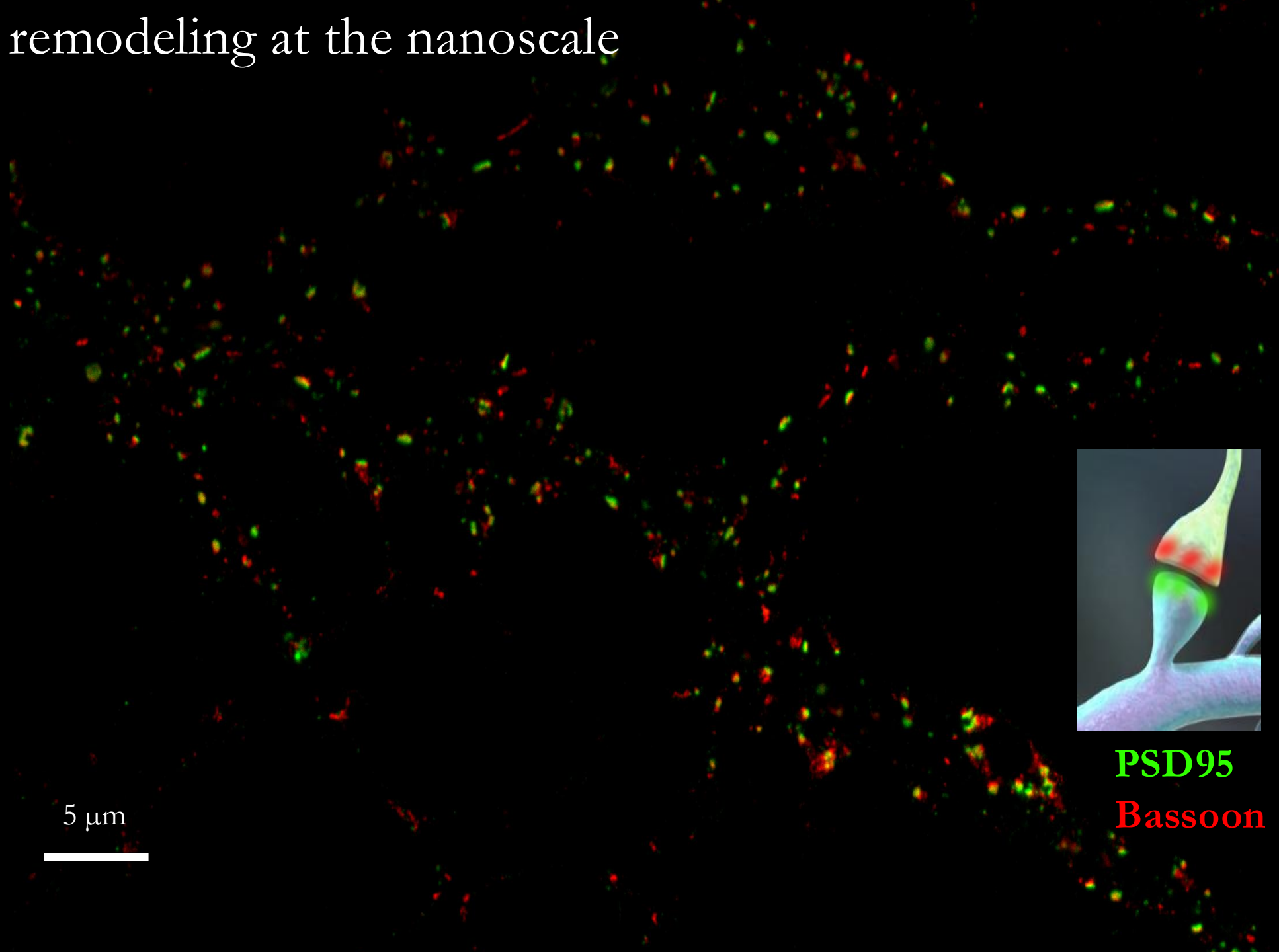
T. Wiesner



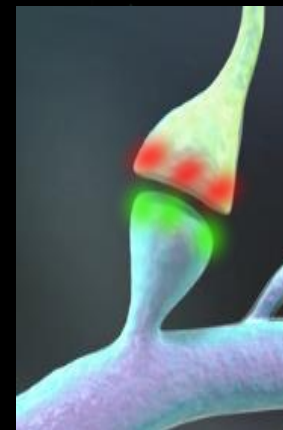
F-actin

Confocal

Synaptic remodeling at the nanoscale



5 μm

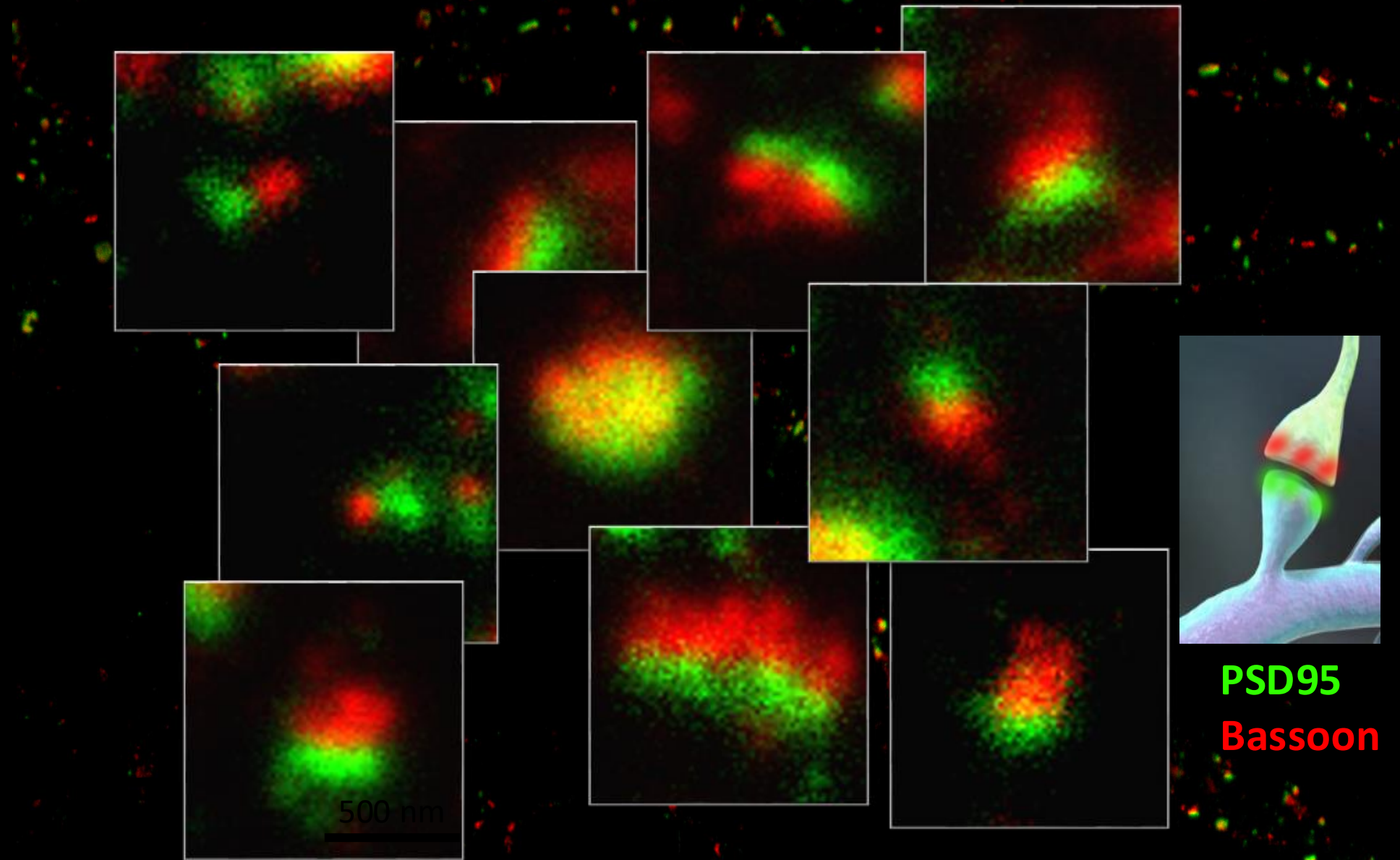


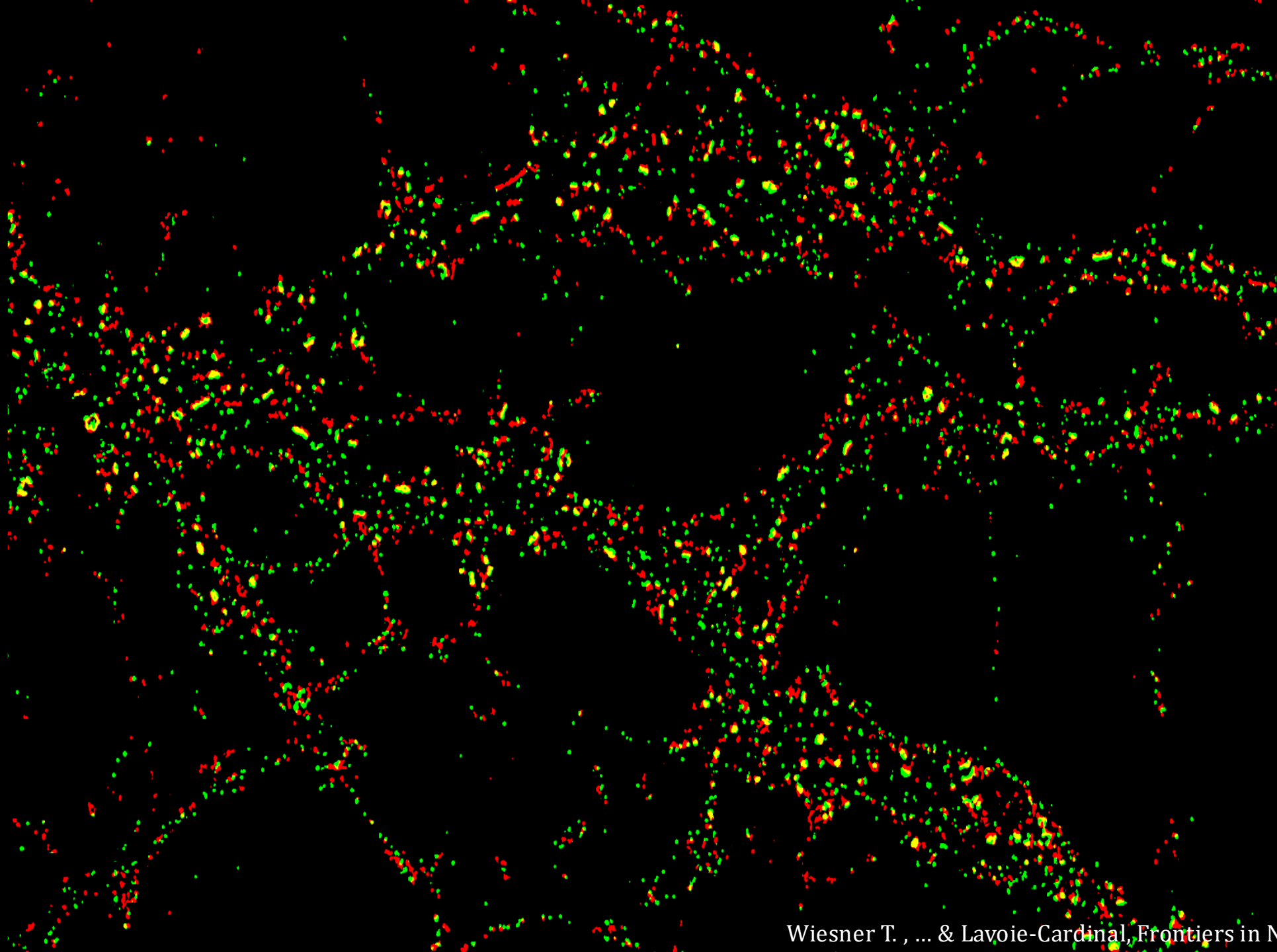
PSD95

Bassoon

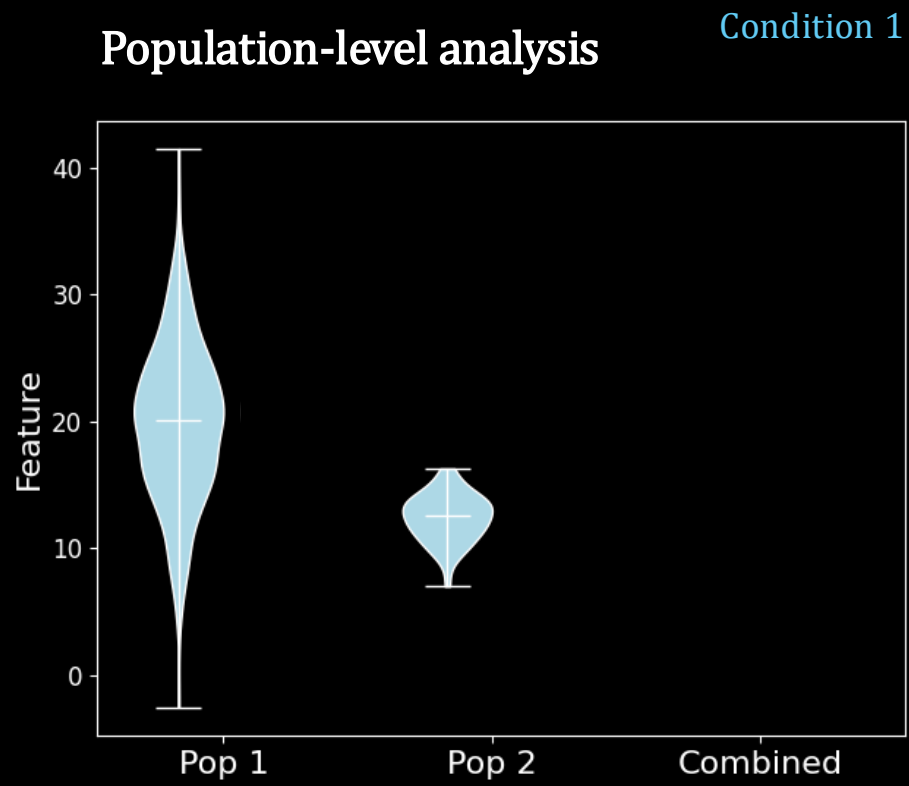
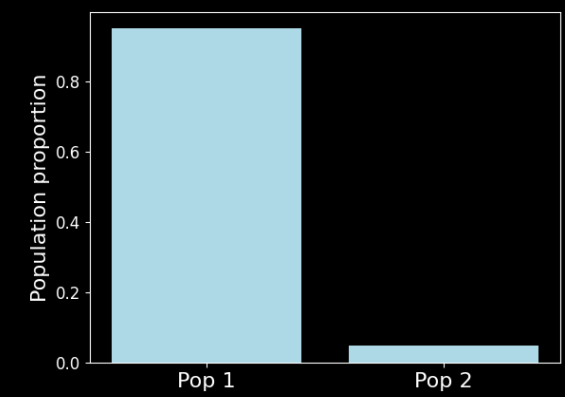
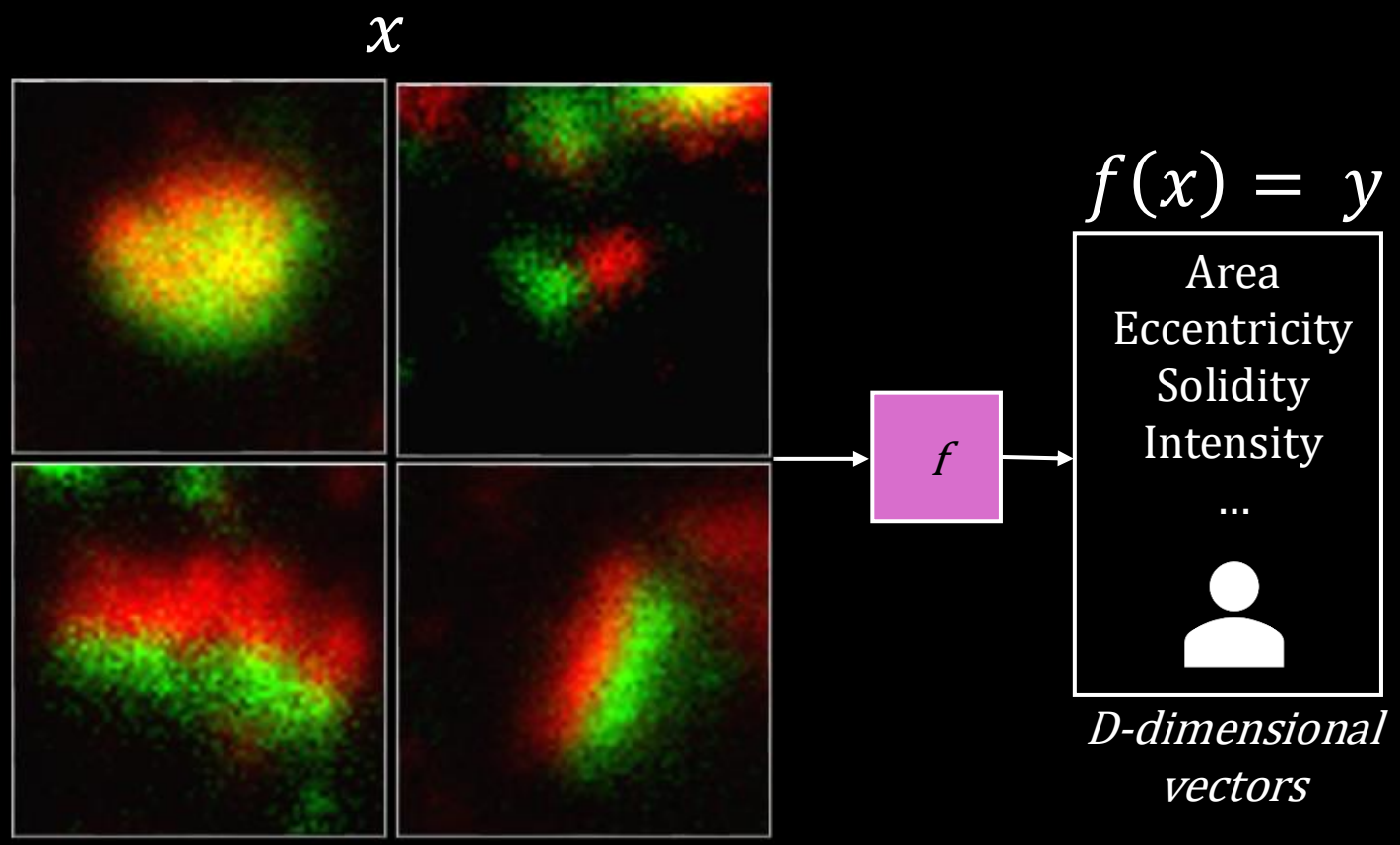
STED

Quantitative analysis of synaptic morphology

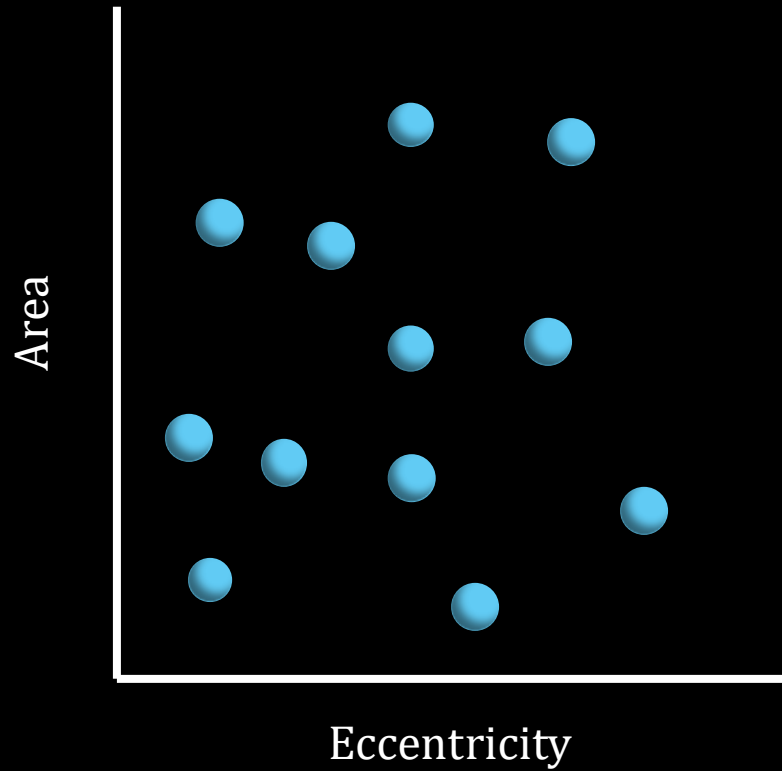
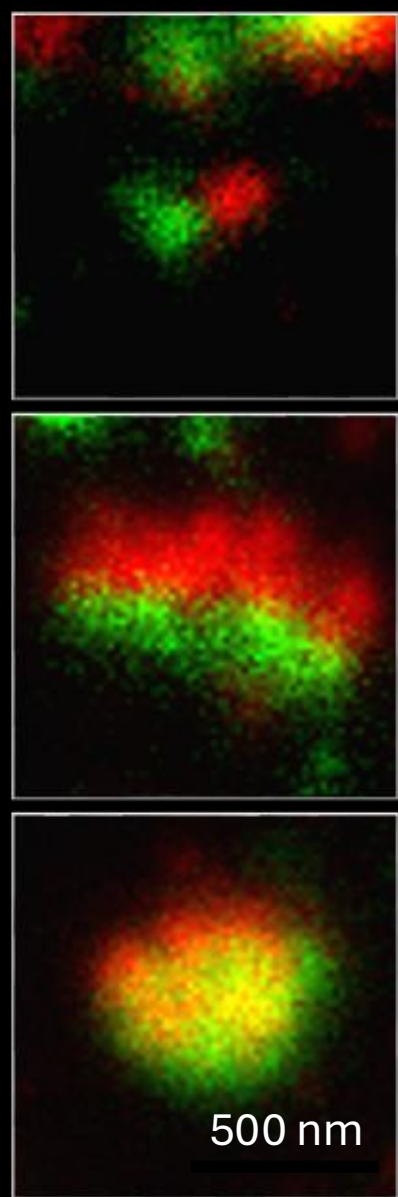




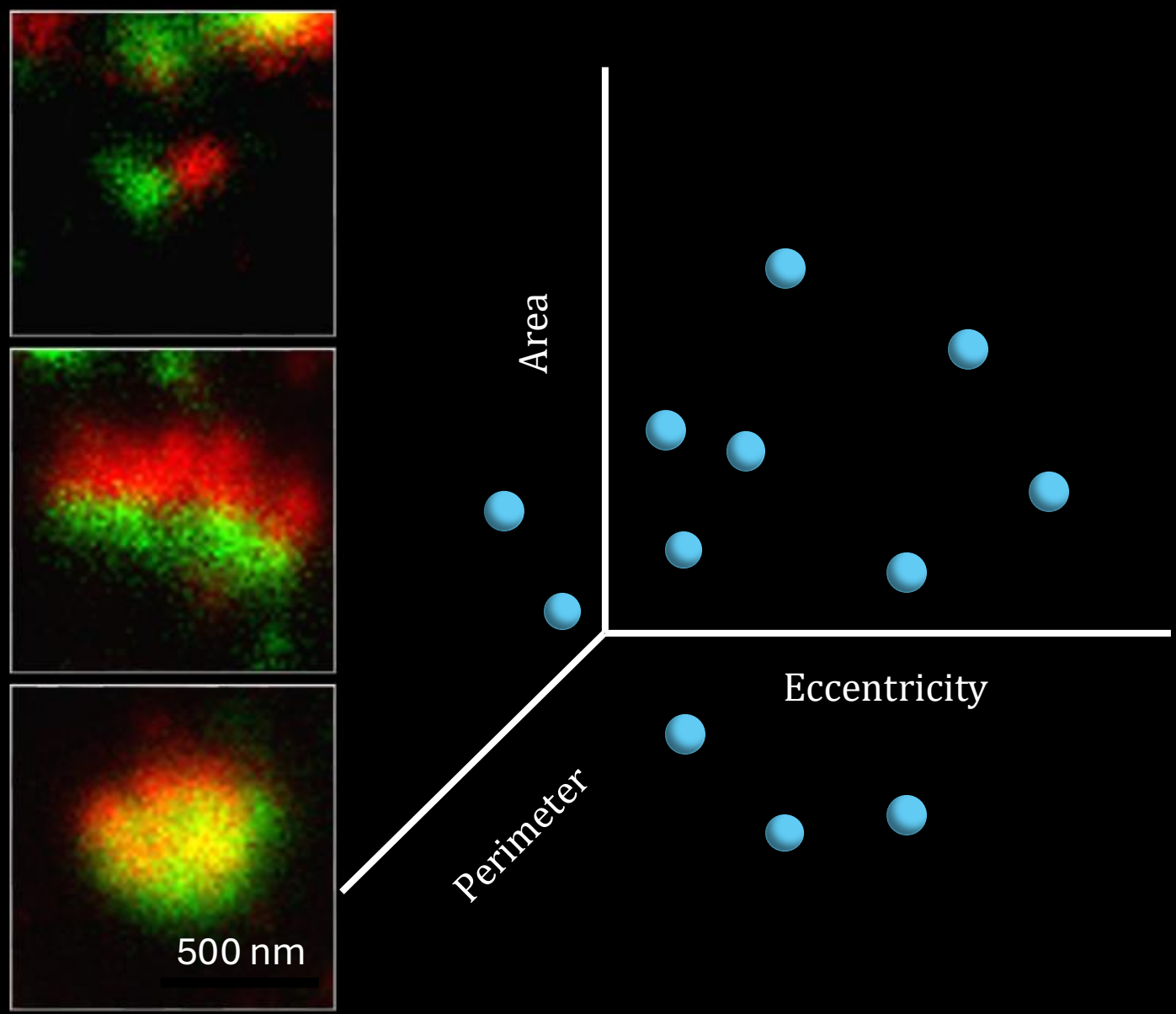
Quantitative analysis of synaptic organization



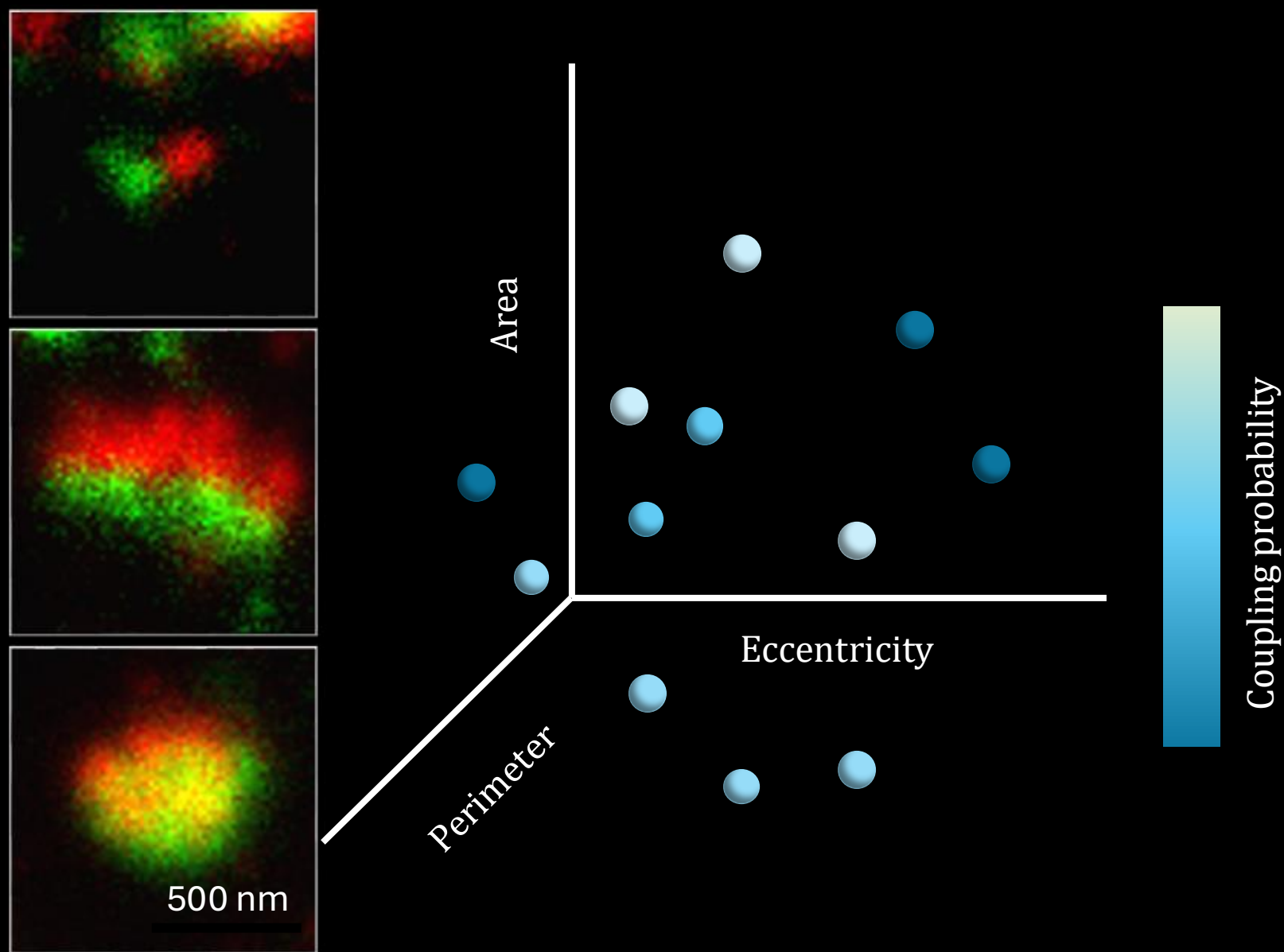
Quantitative analysis of synaptic organization



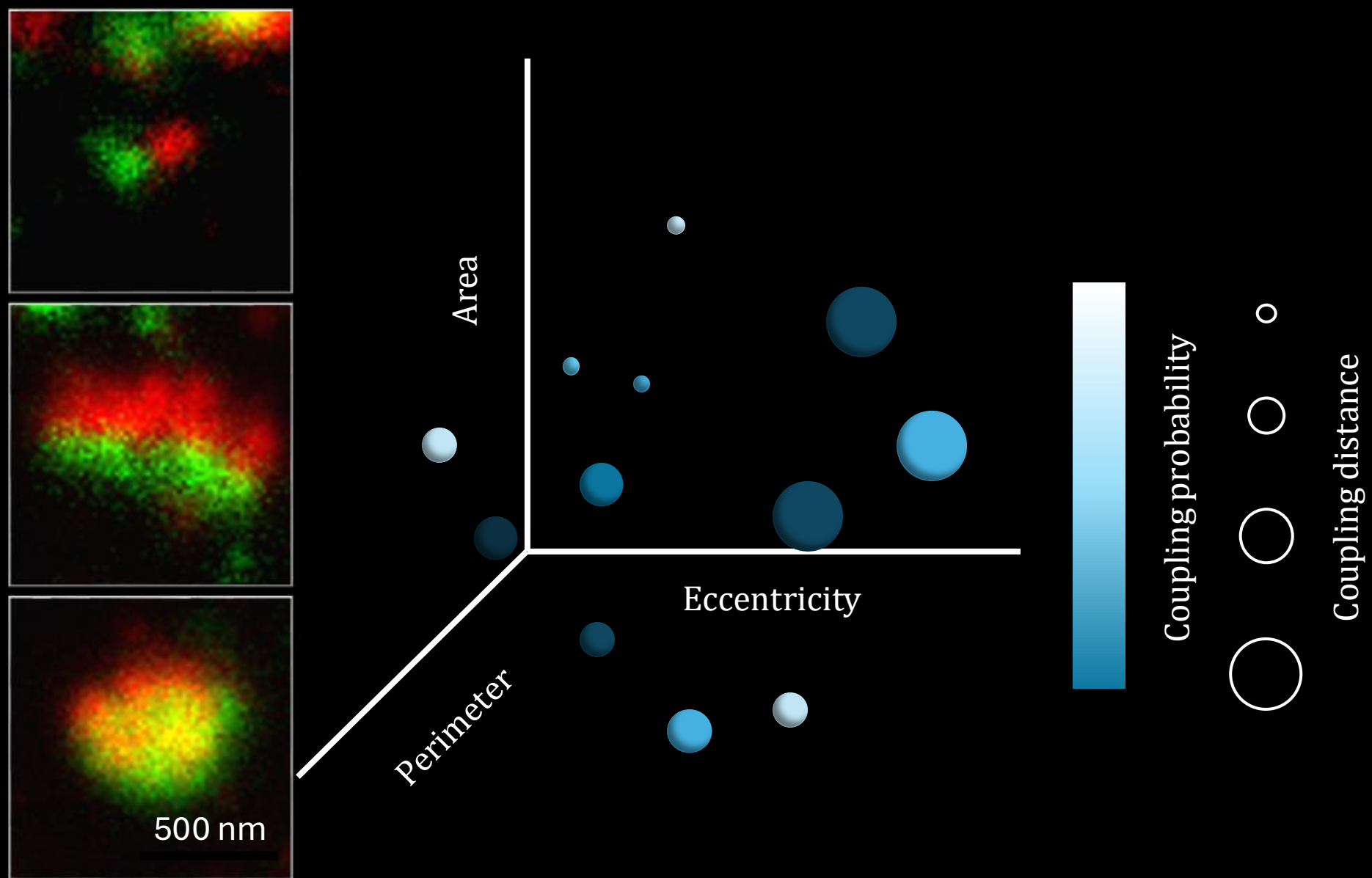
Quantitative analysis of synaptic organization



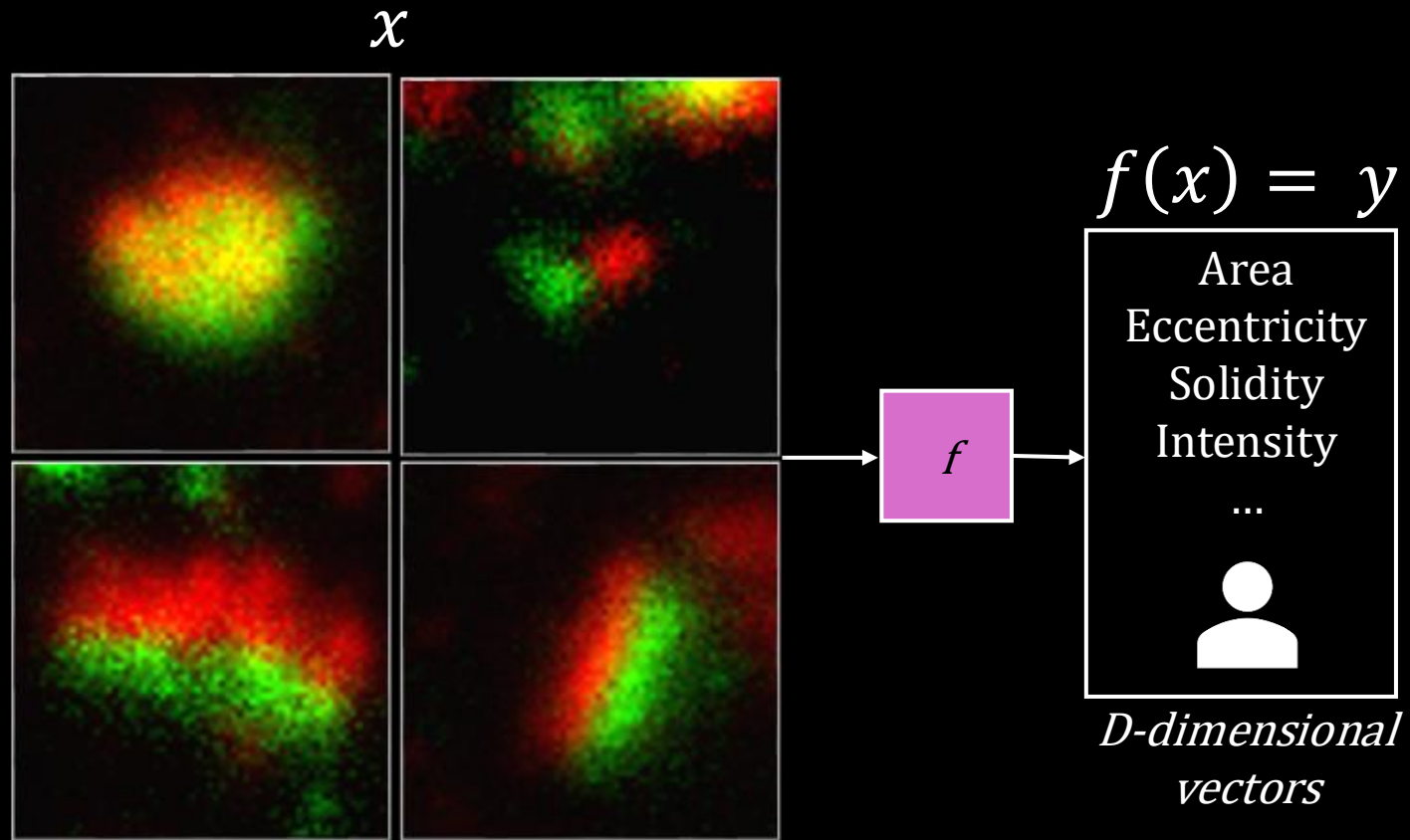
Quantitative analysis of synaptic organization



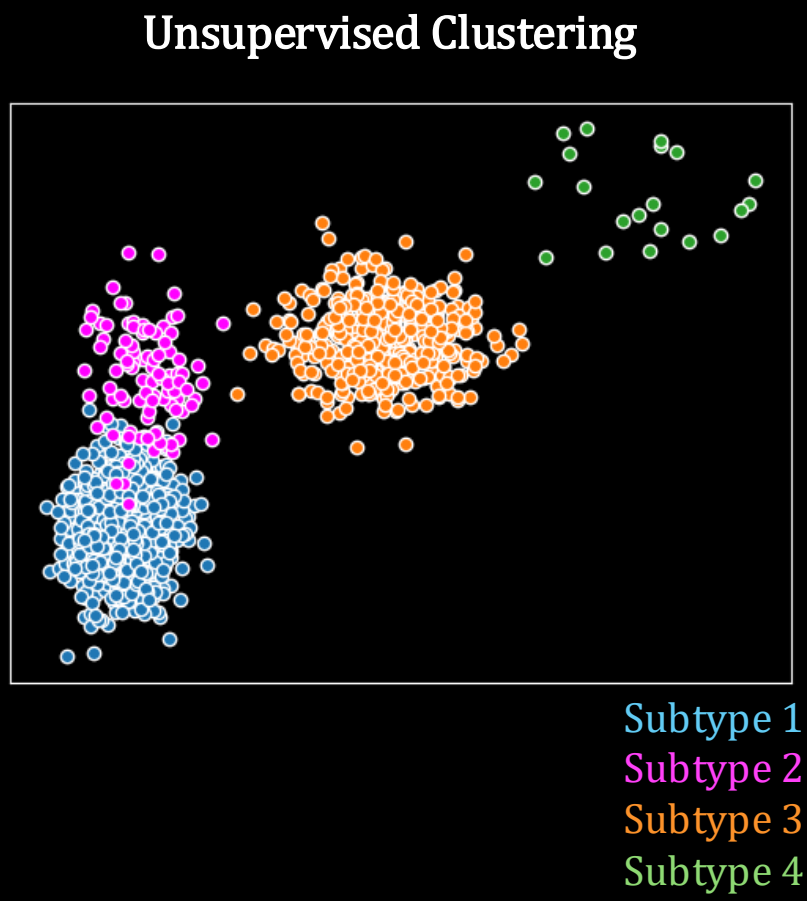
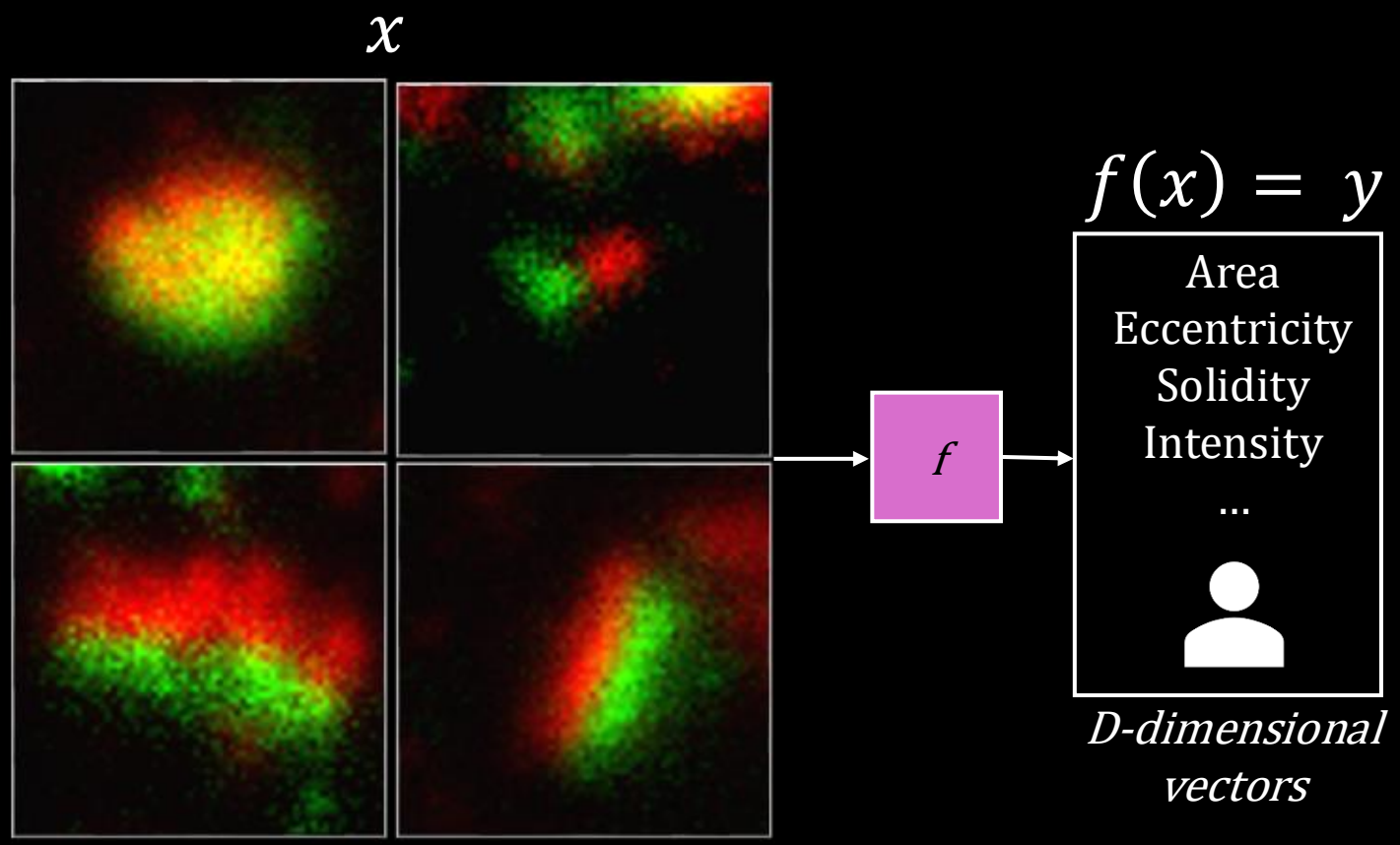
Quantitative analysis of synaptic organization

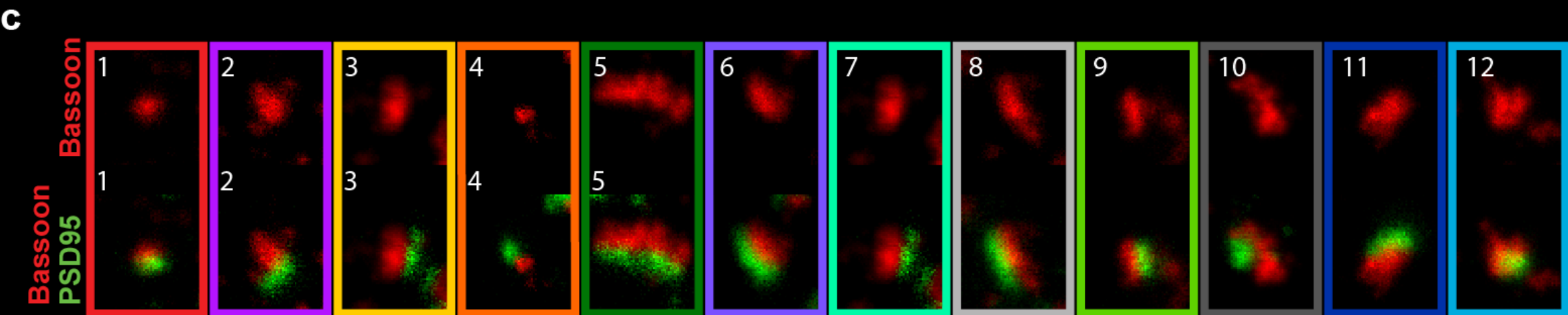
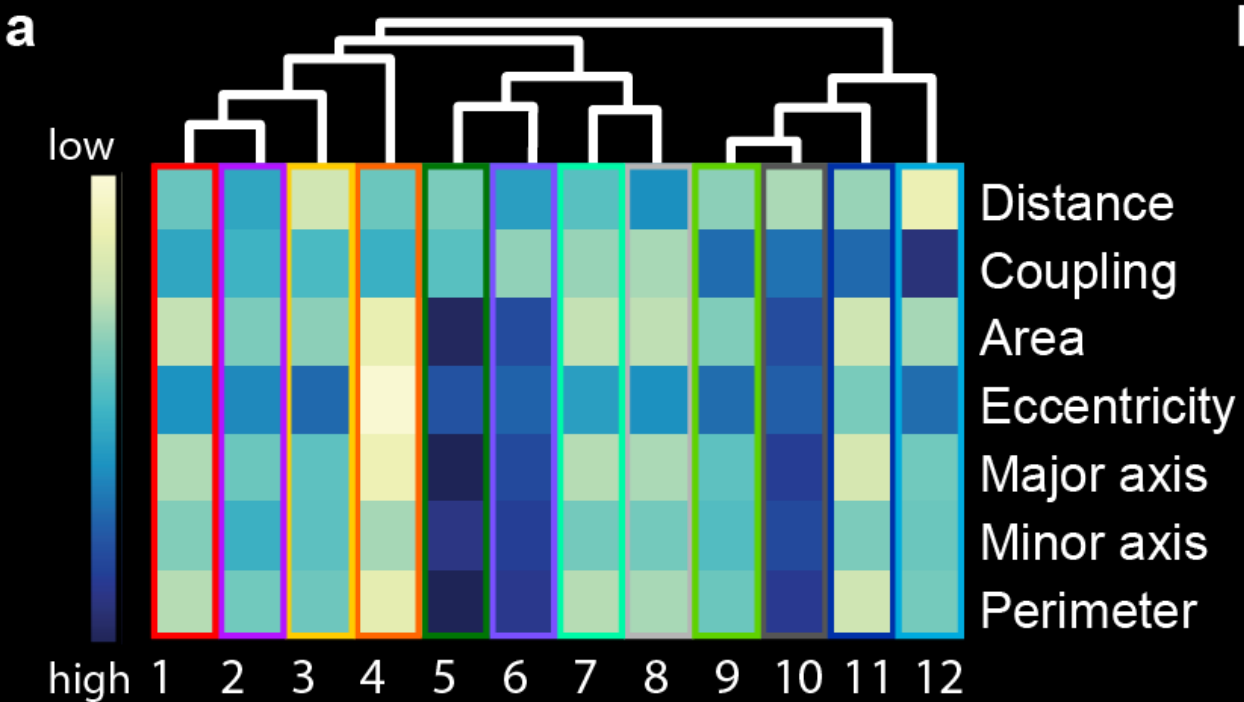


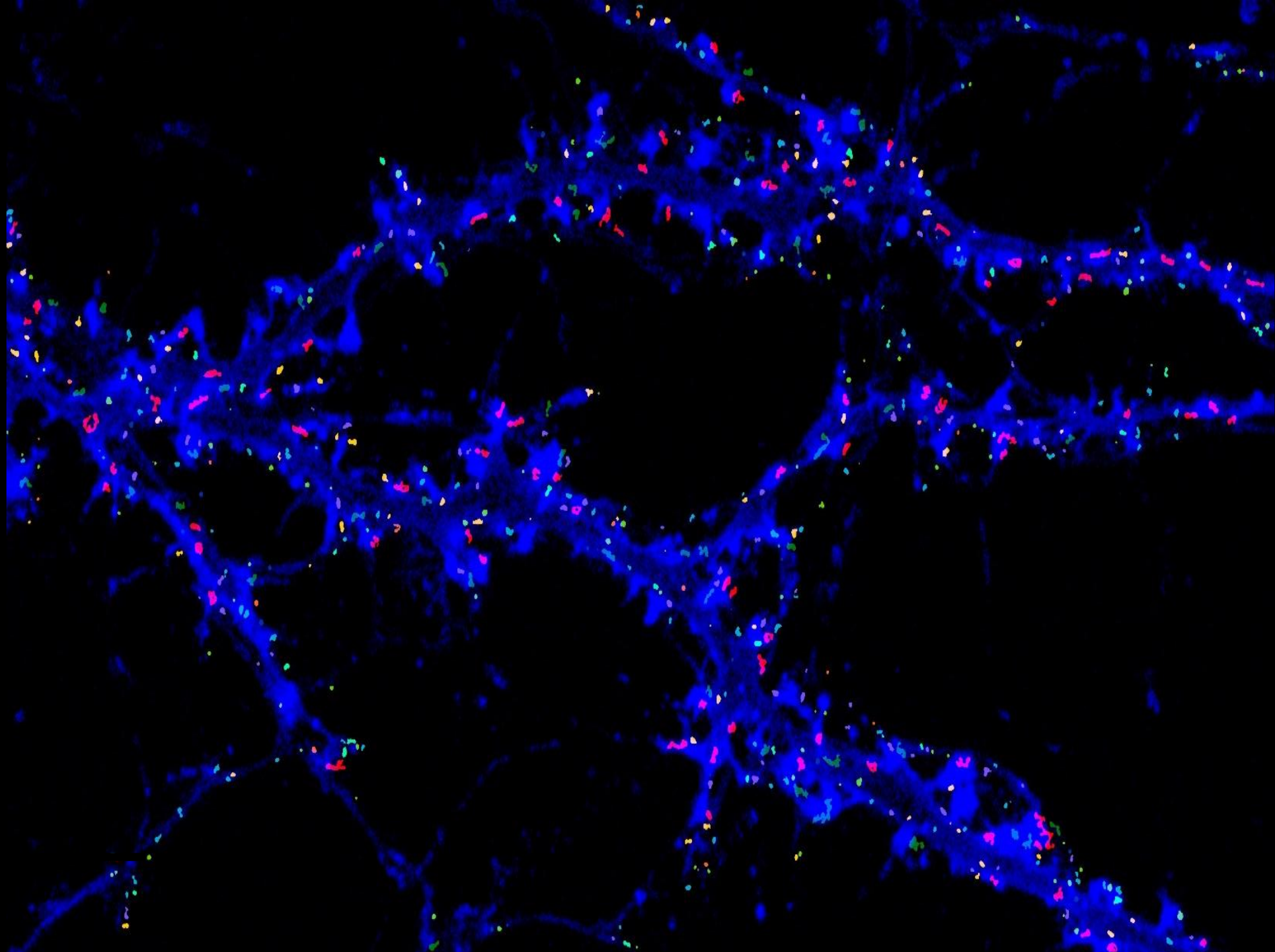
Multidimensional analysis

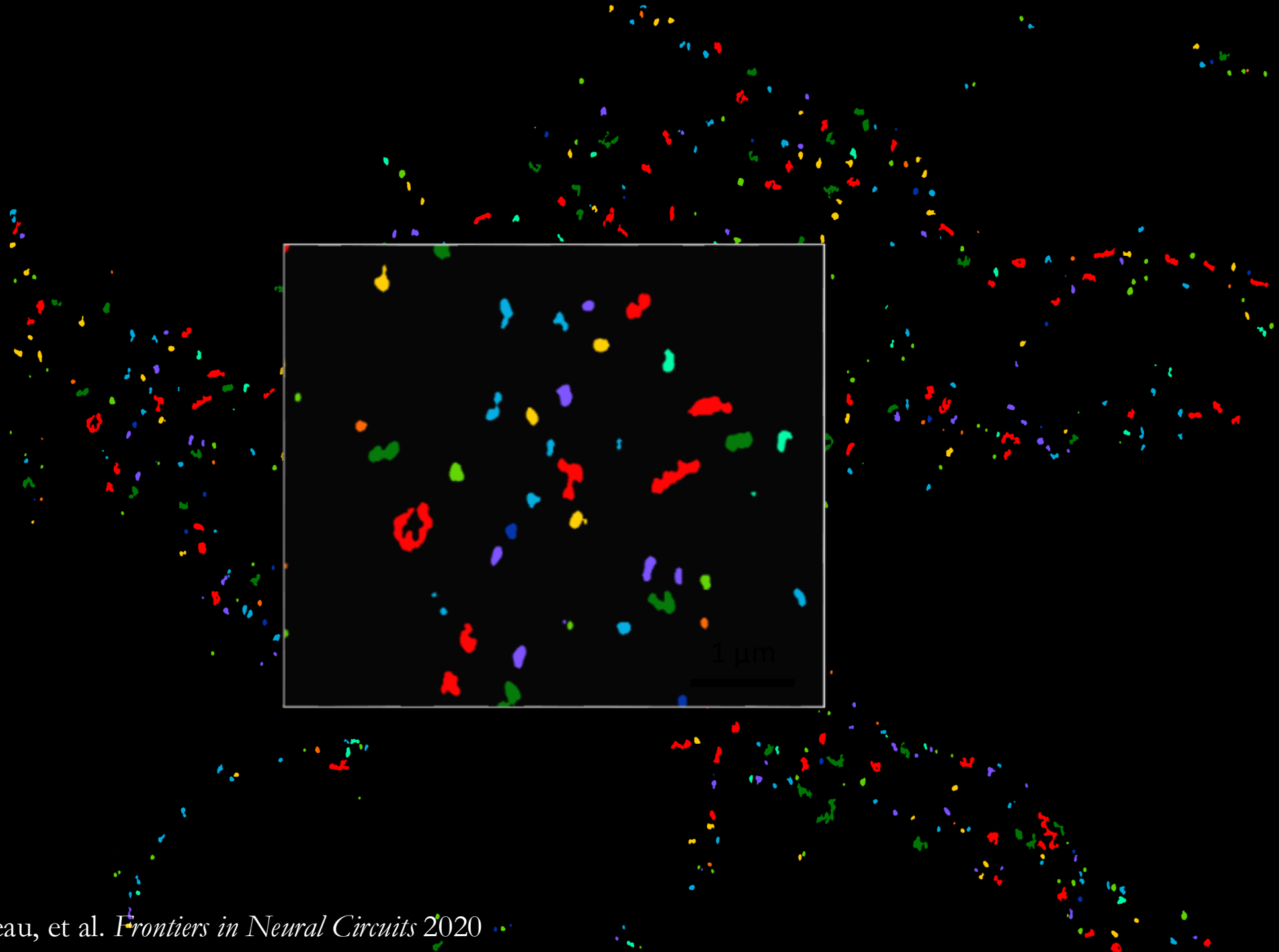


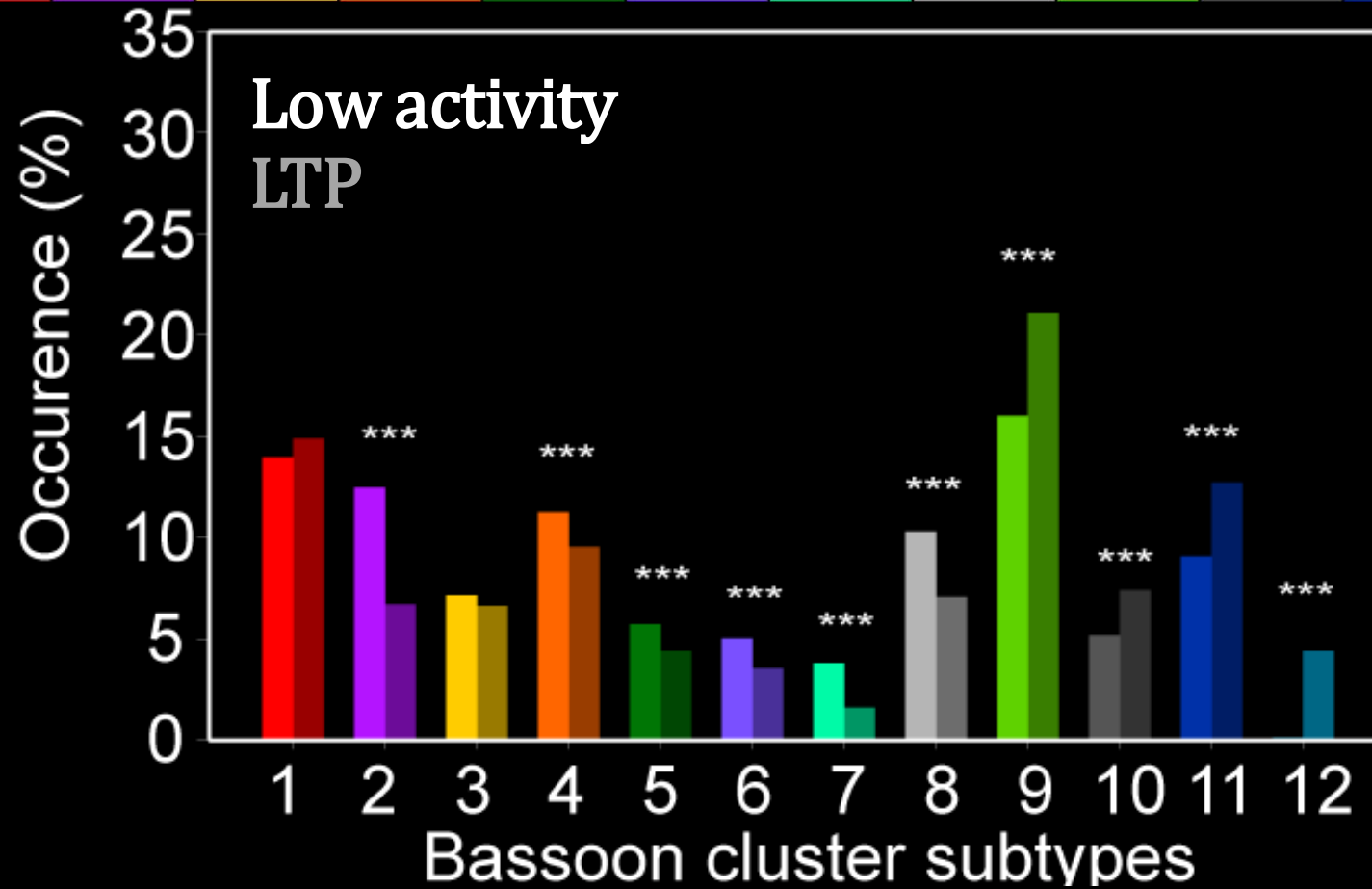
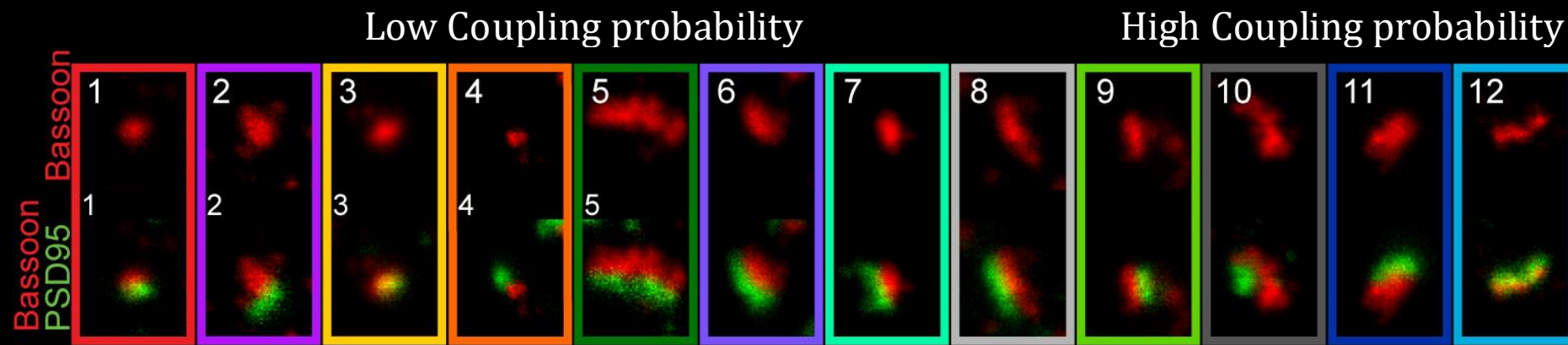
Multidimensional analysis







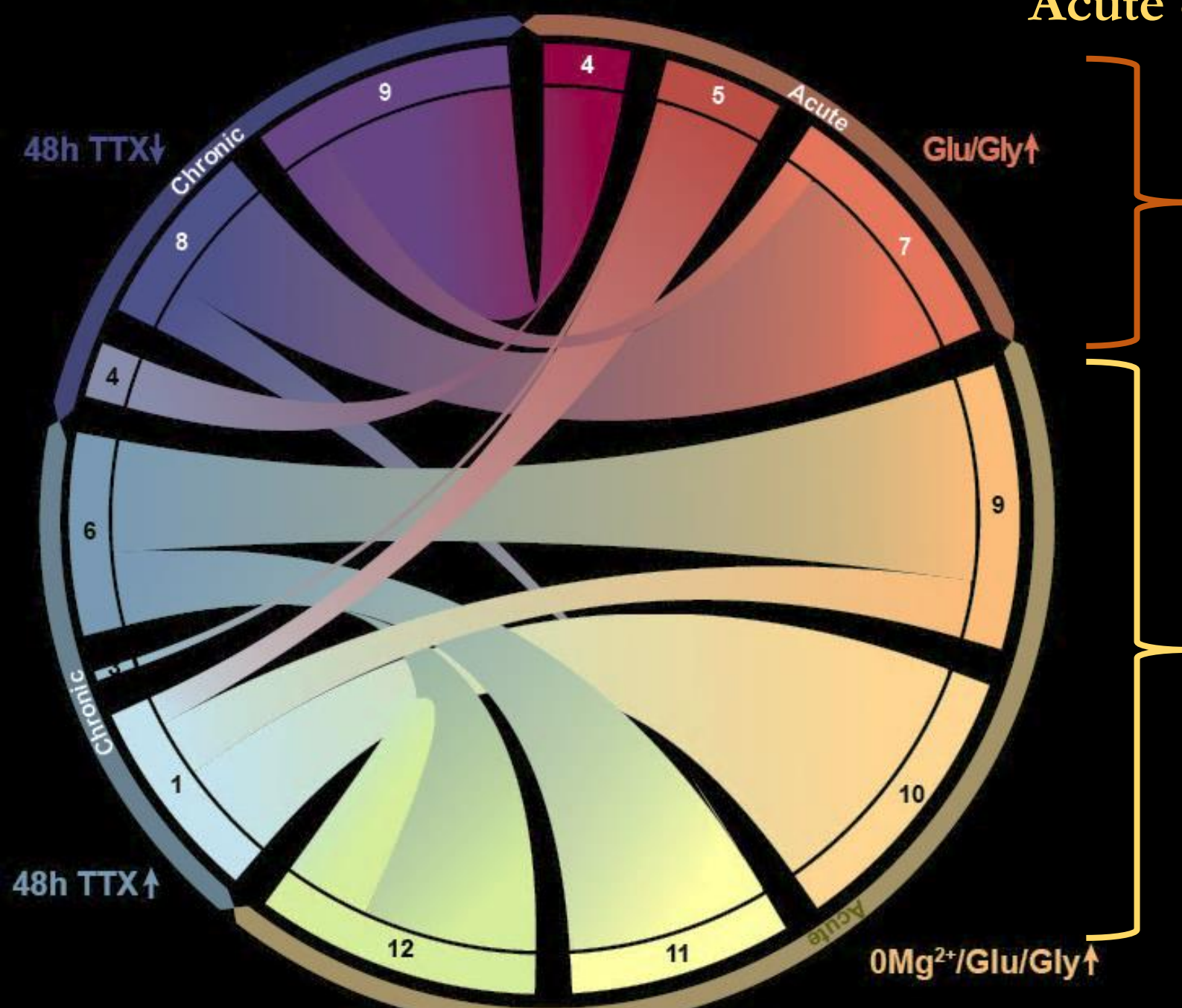




Chronic TTX

Bassoon

Acute cLTP/cLTD



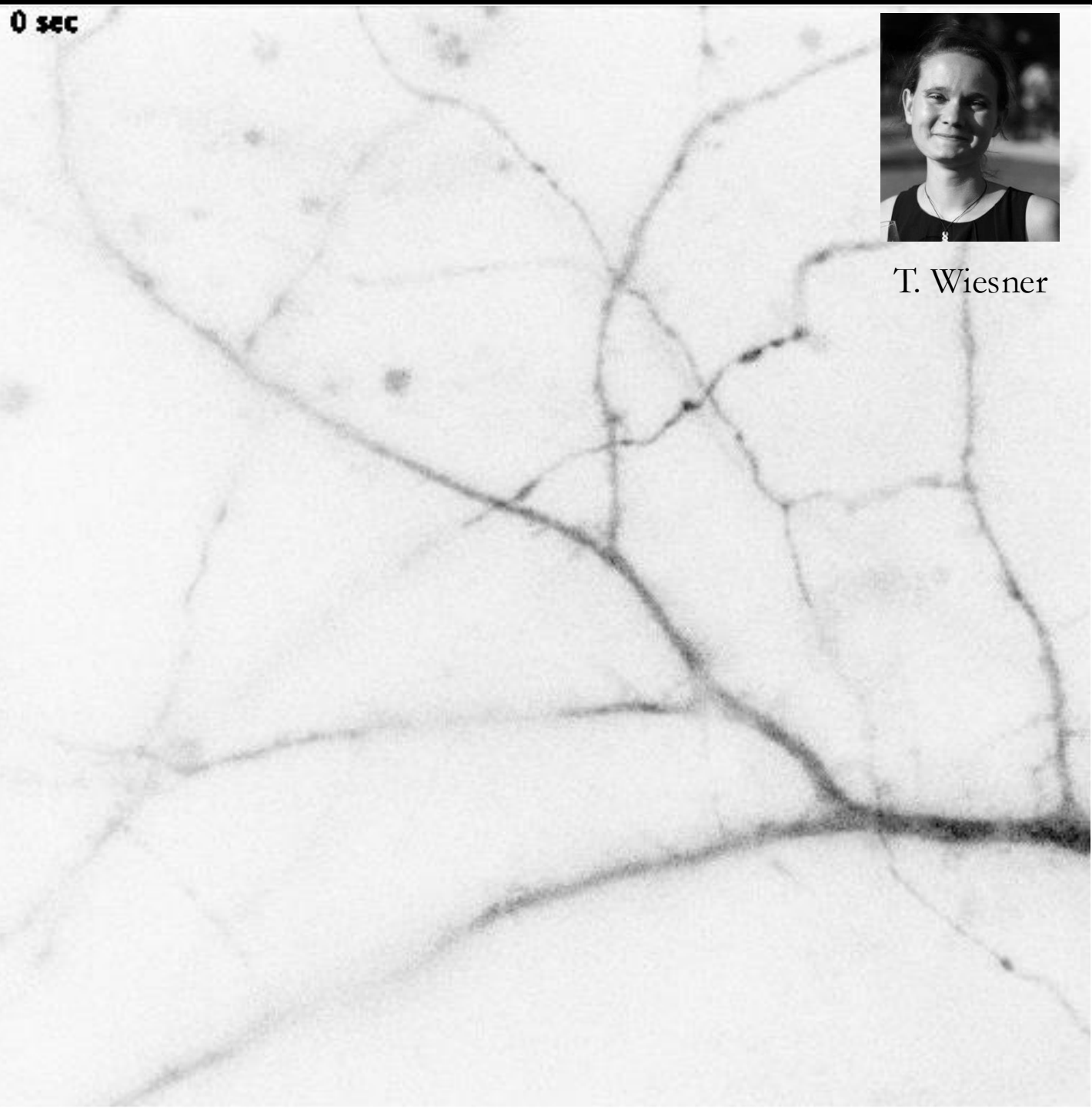


Frédéric Beaupré

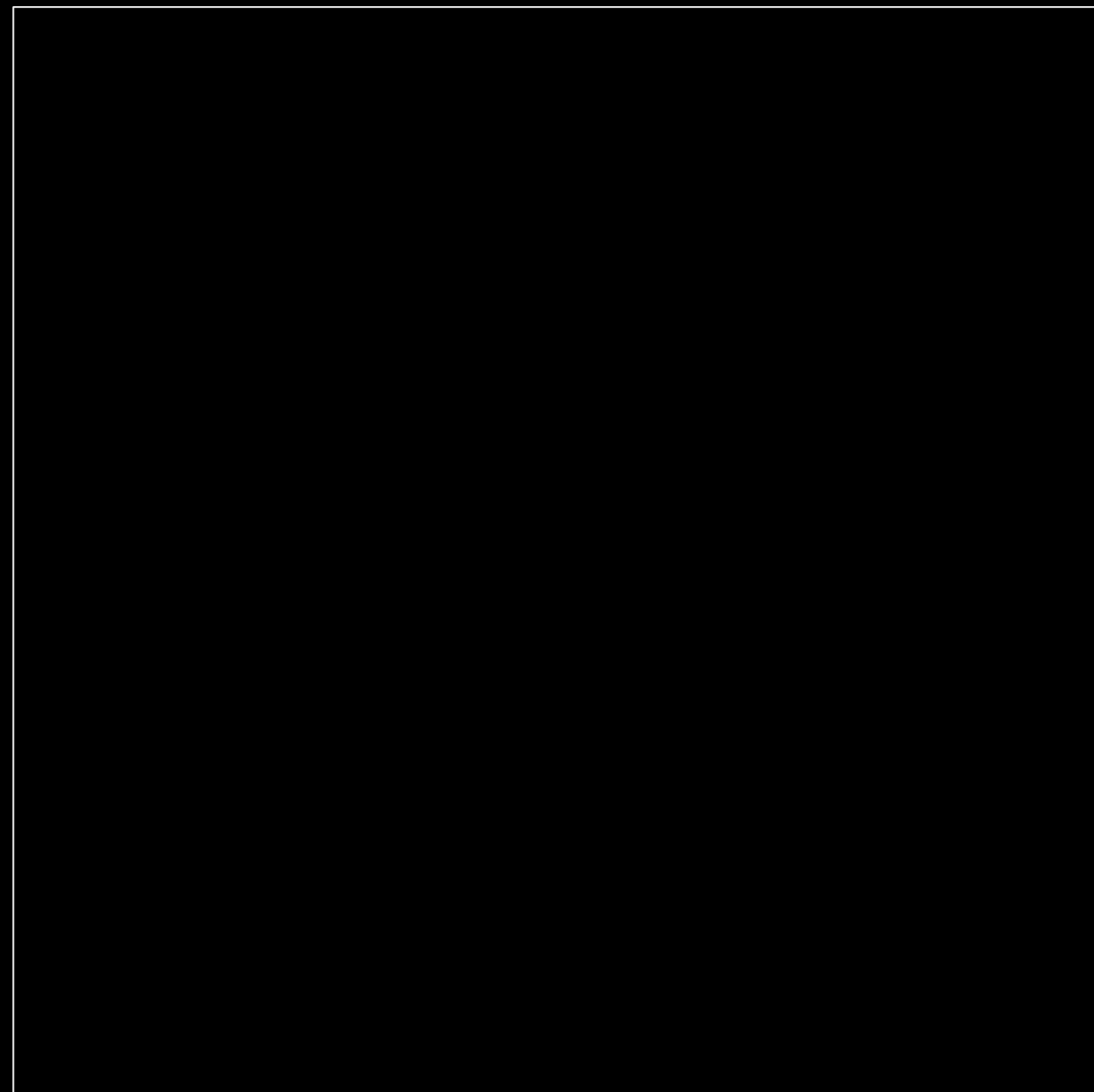
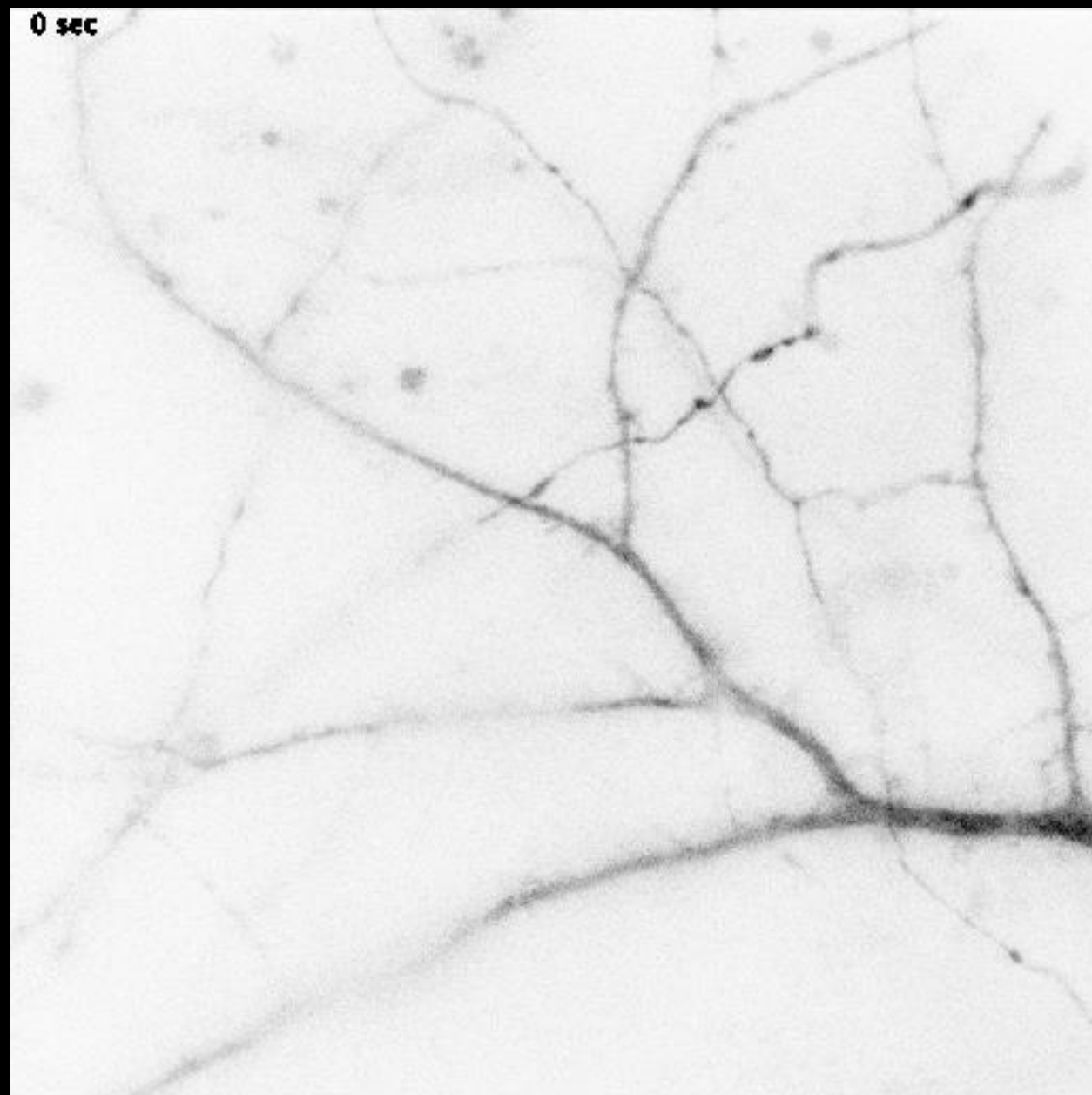


Anthony Bilodeau

Quantitative Analysis of Miniature Synaptic Calcium Transients (mSCTs)

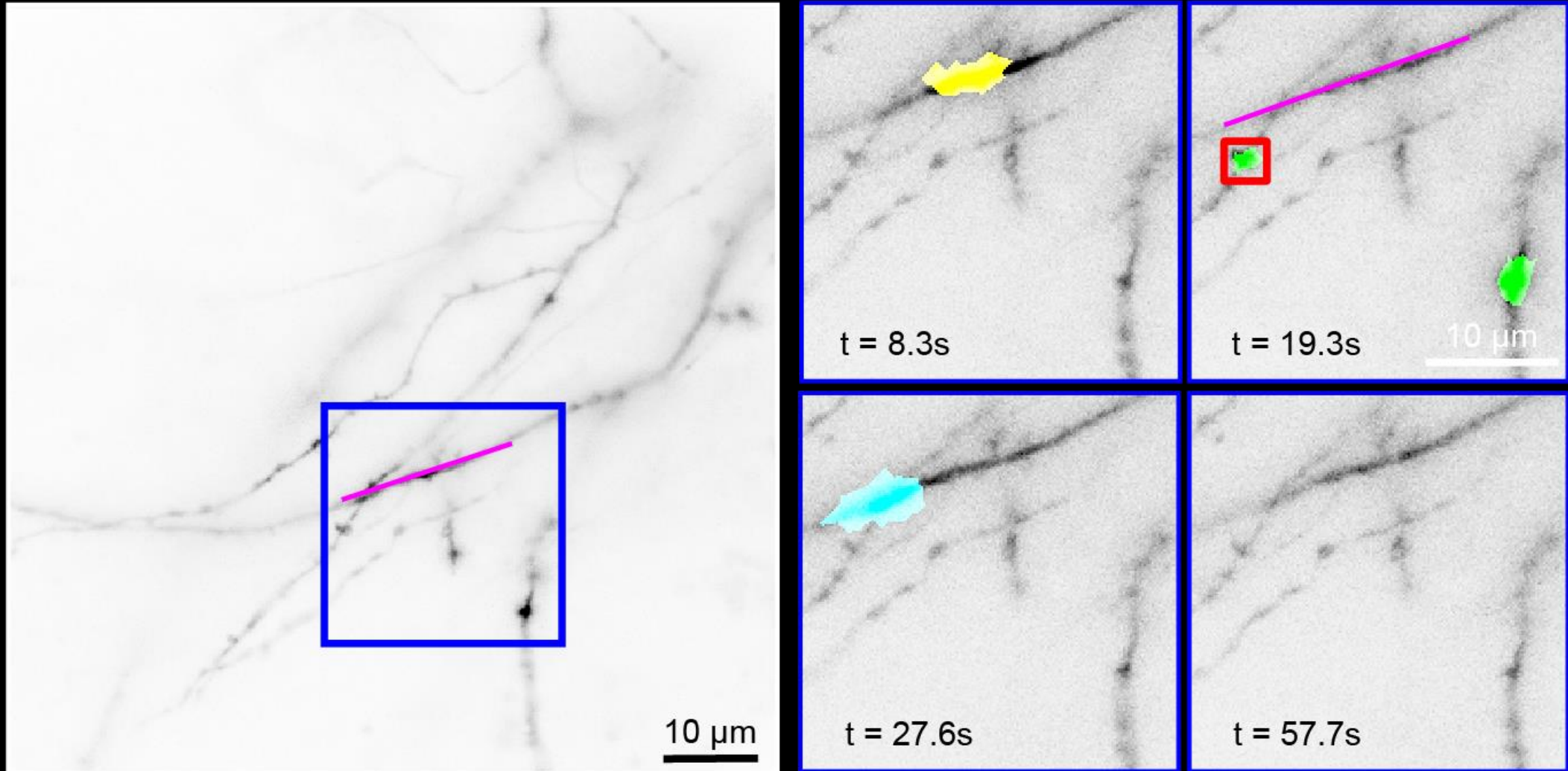


T. Wiesner

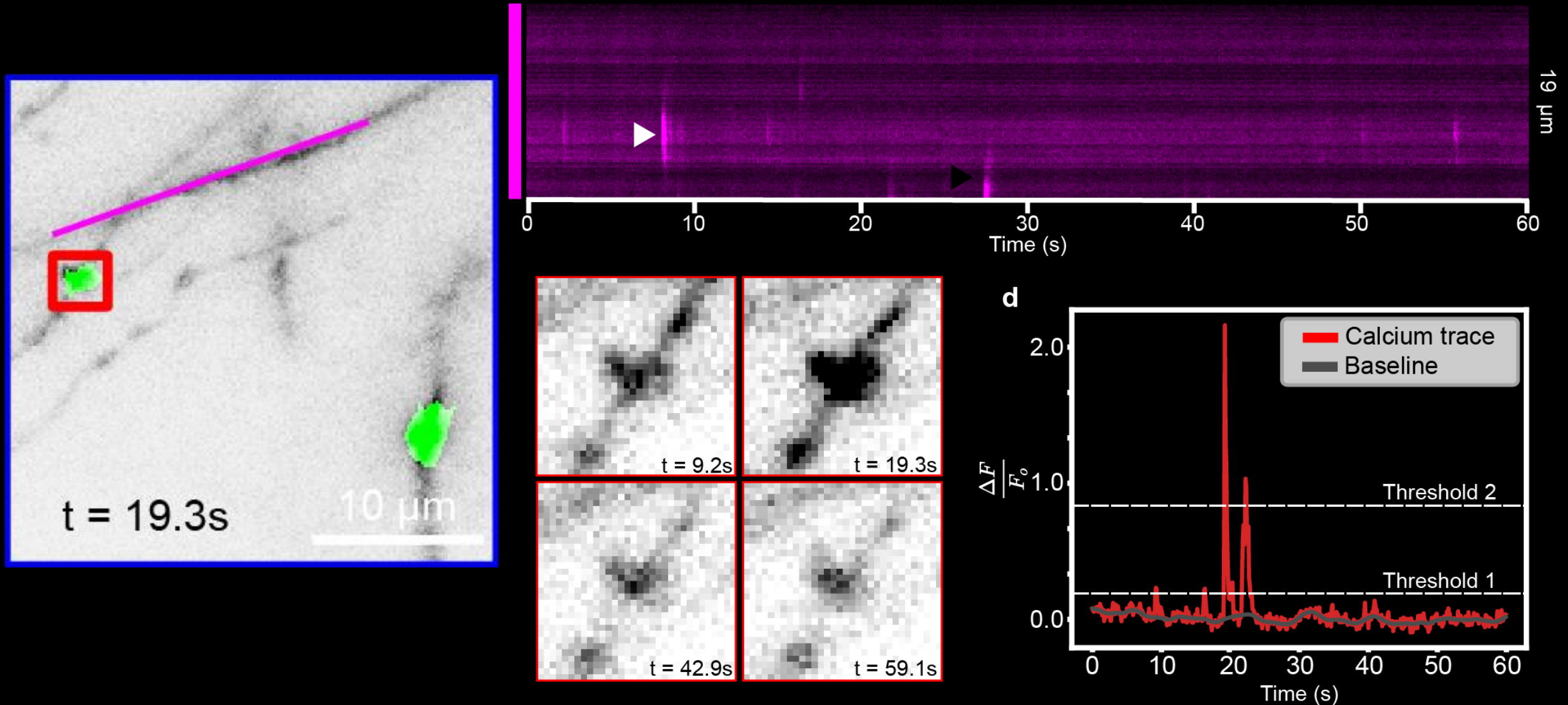


Quantitative Analysis of Miniature Synaptic Calcium Transients

a

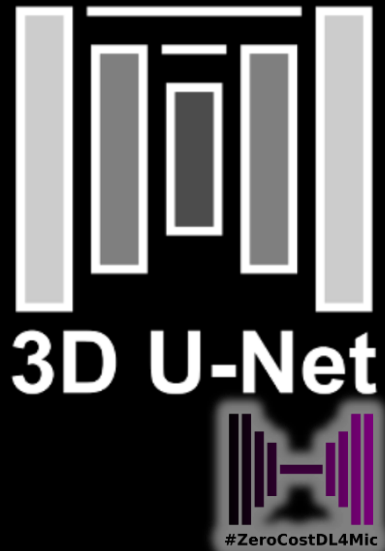
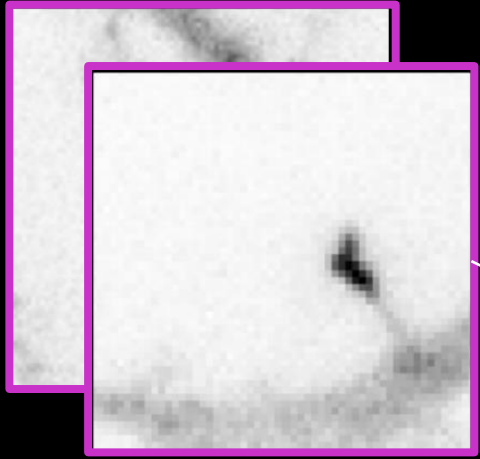


Quantitative Analysis of Miniature Synaptic Calcium Transients

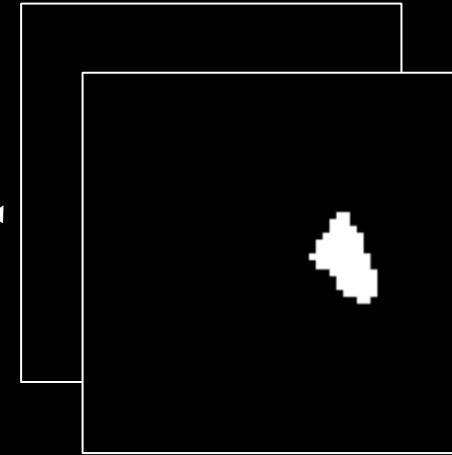


Training a deep neural network for the detection and segmentation of mSCTs

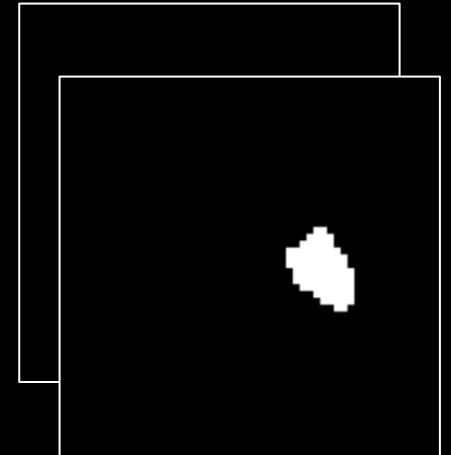
Positives instances



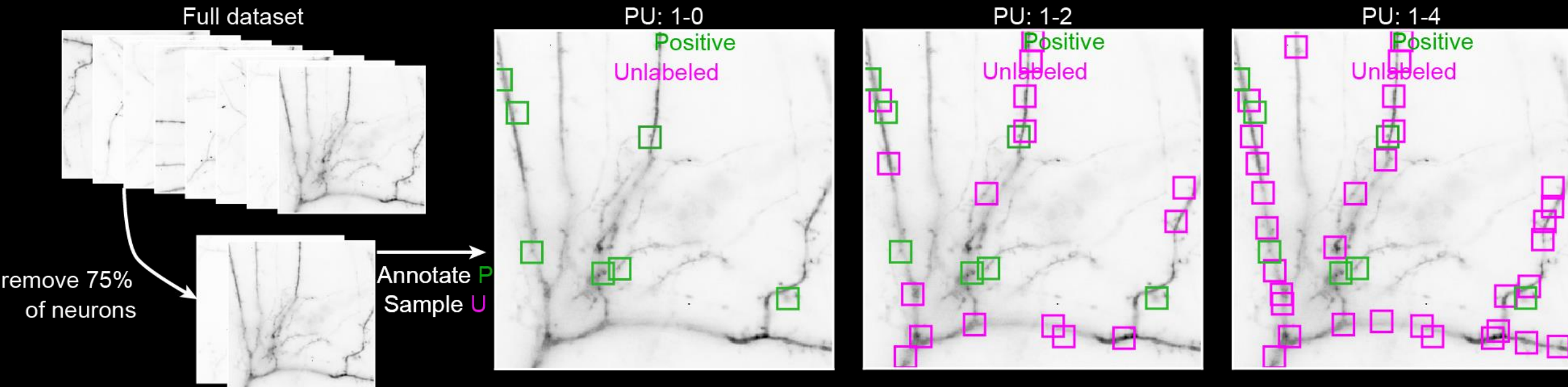
Prediction



Ground truth

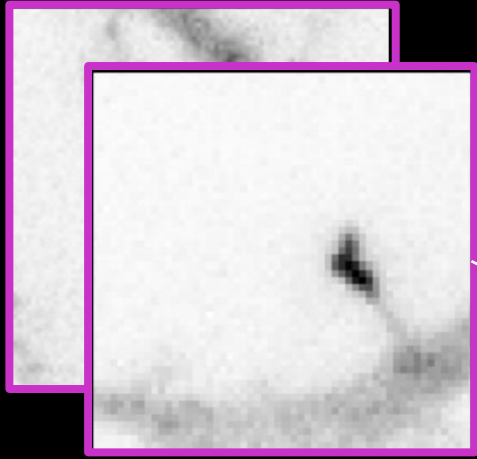


Positive Unlabeled learning

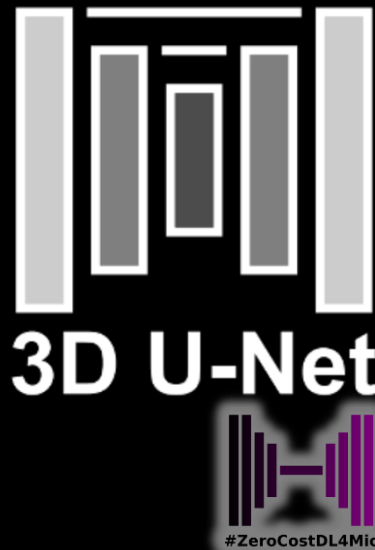
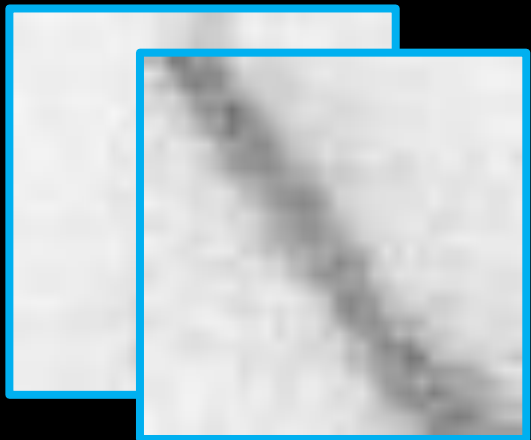


Training a deep segmentation network

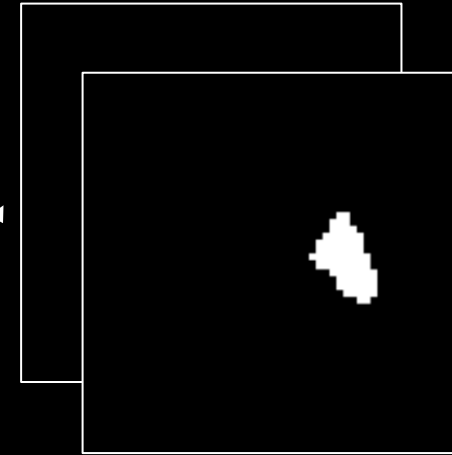
Positives instances



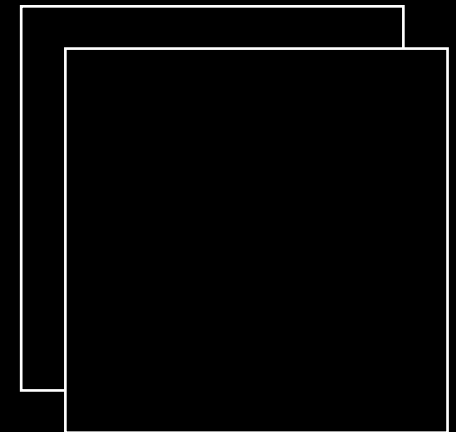
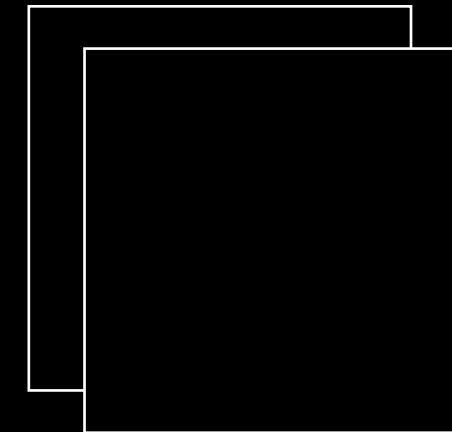
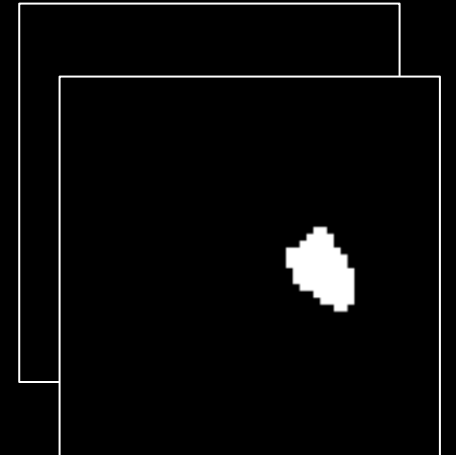
Unlabeled instances



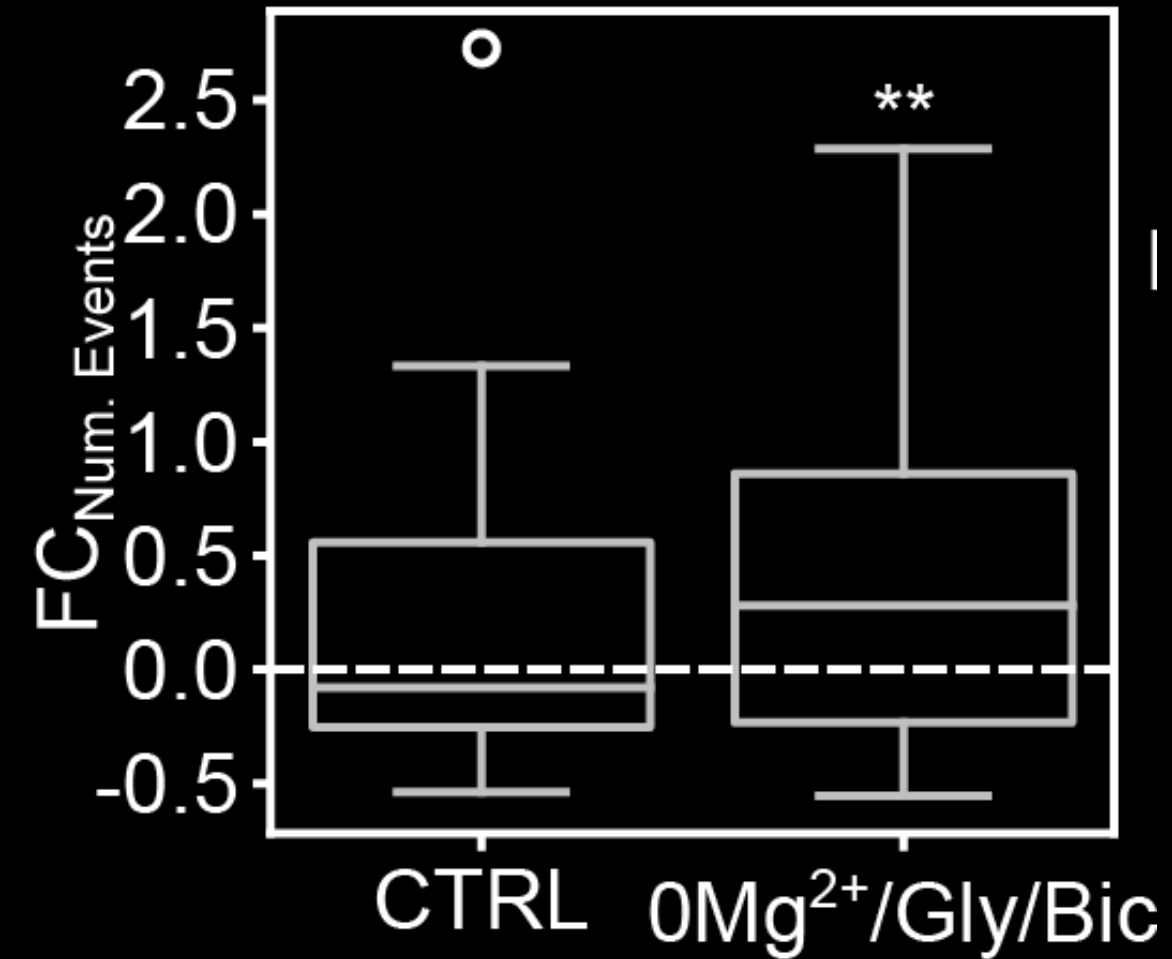
Prediction



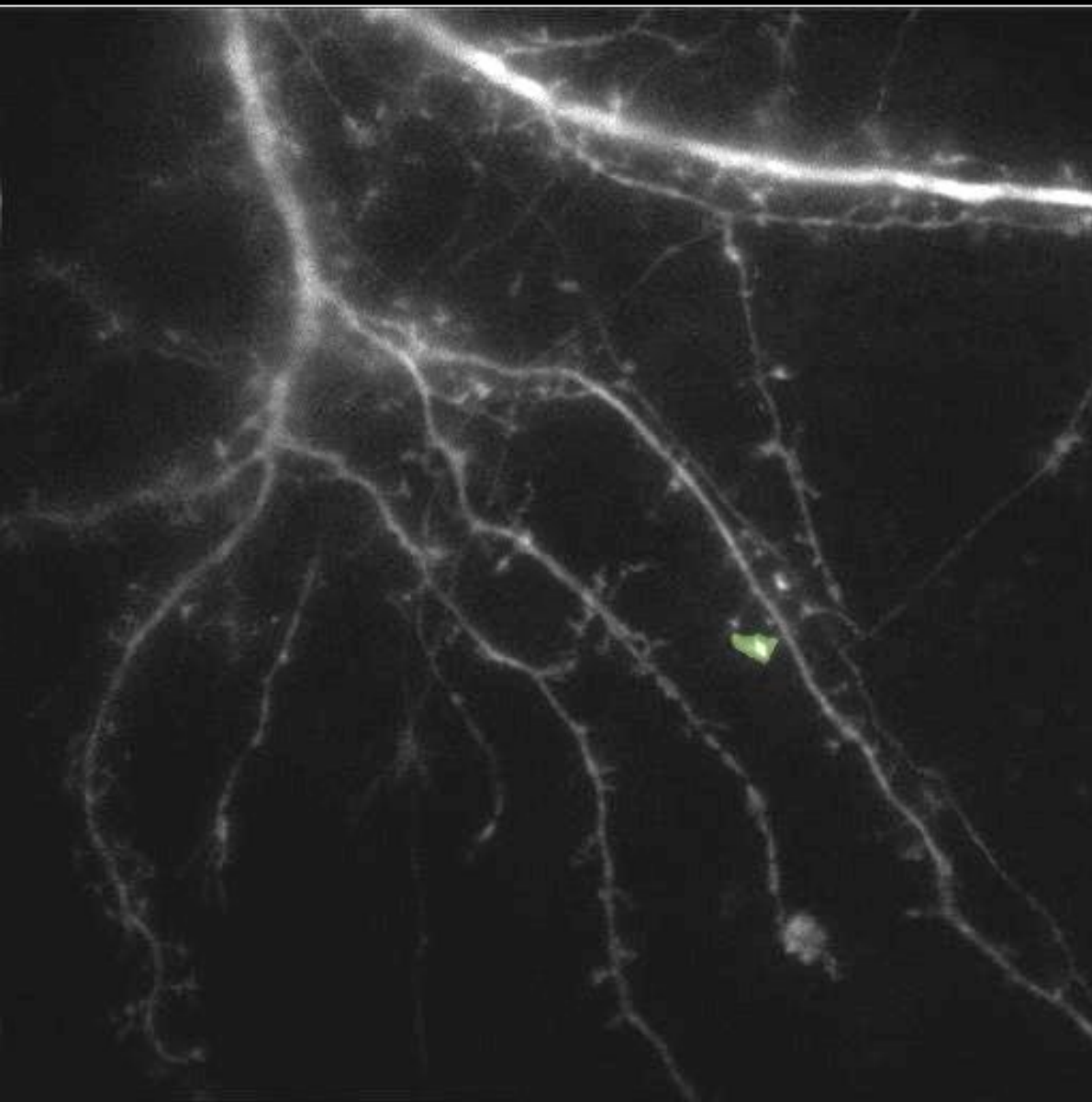
Ground truth



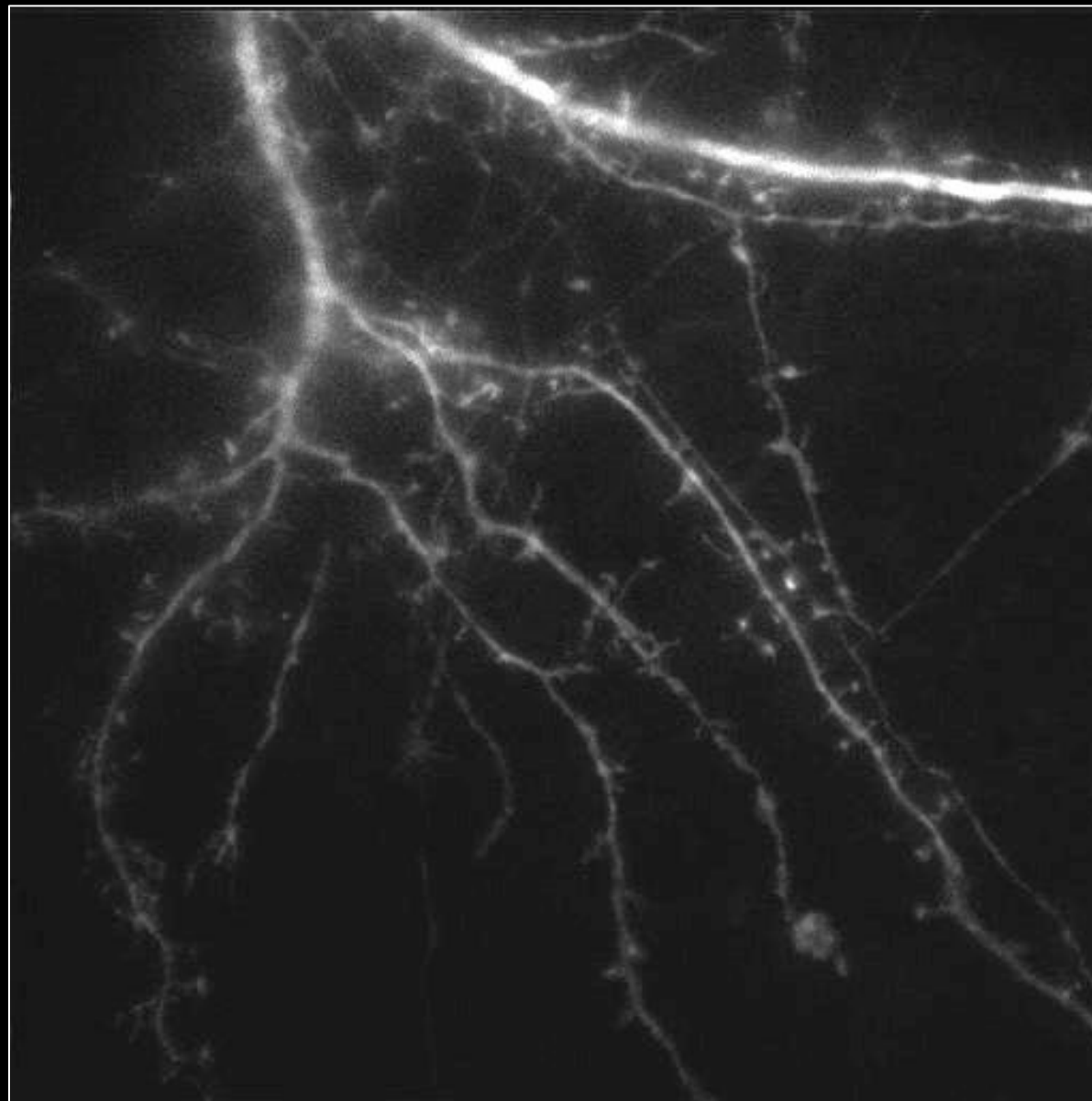
Synaptic potentiation is supported by an increase in the number of detected calcium transients



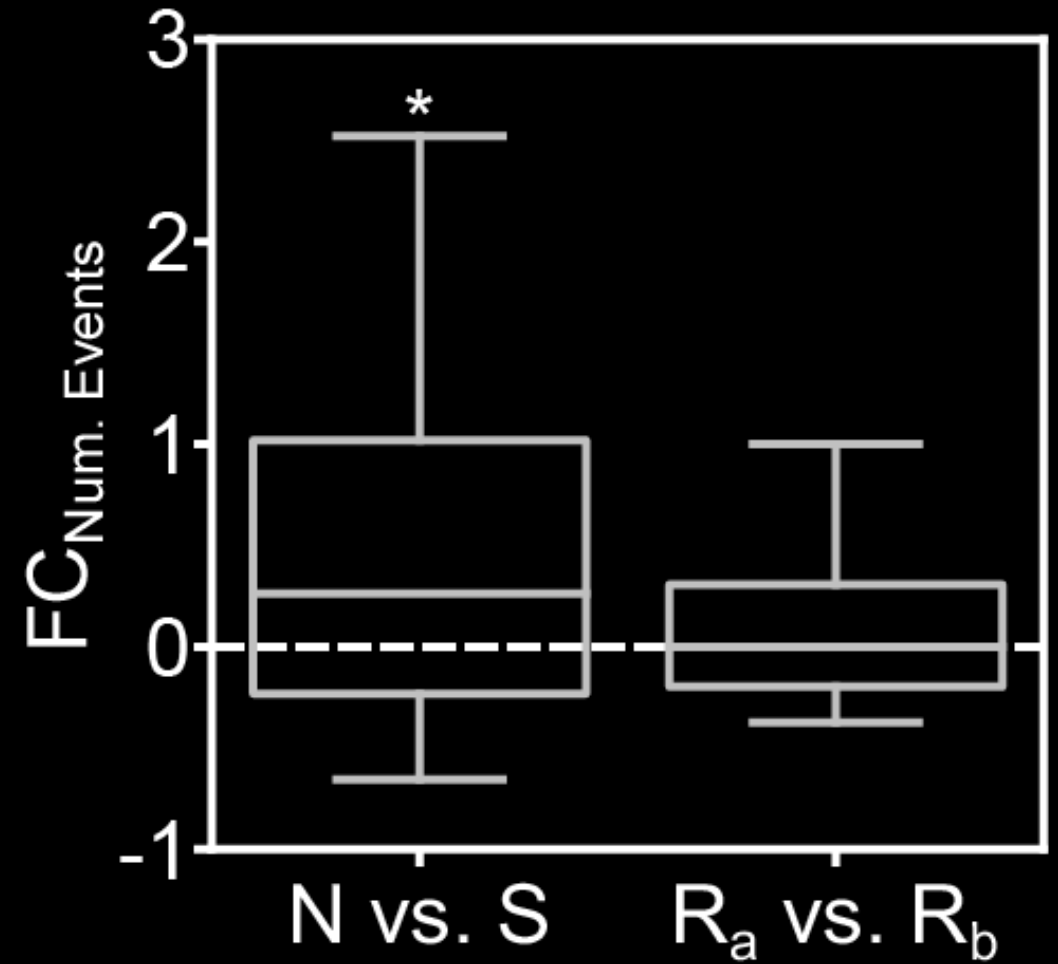
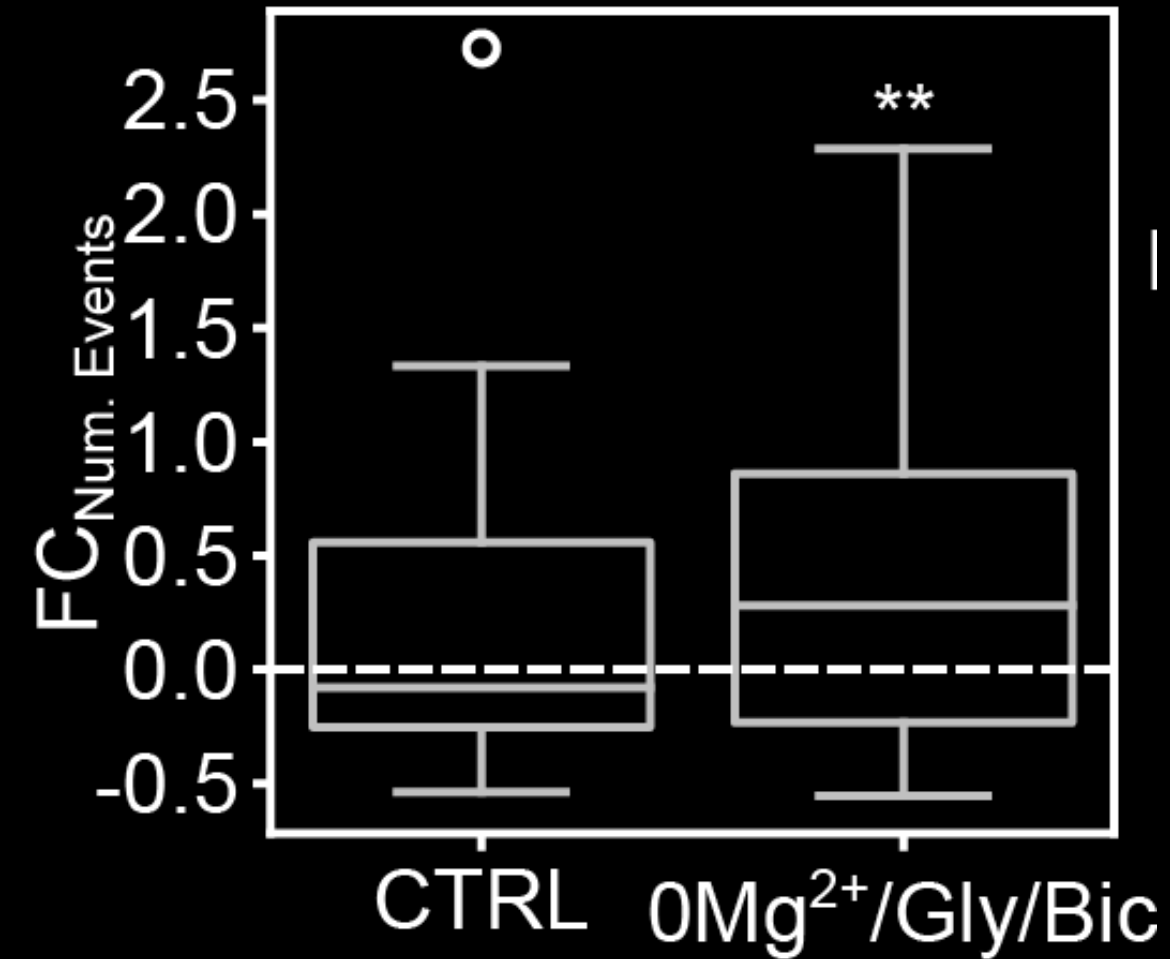
Baseline



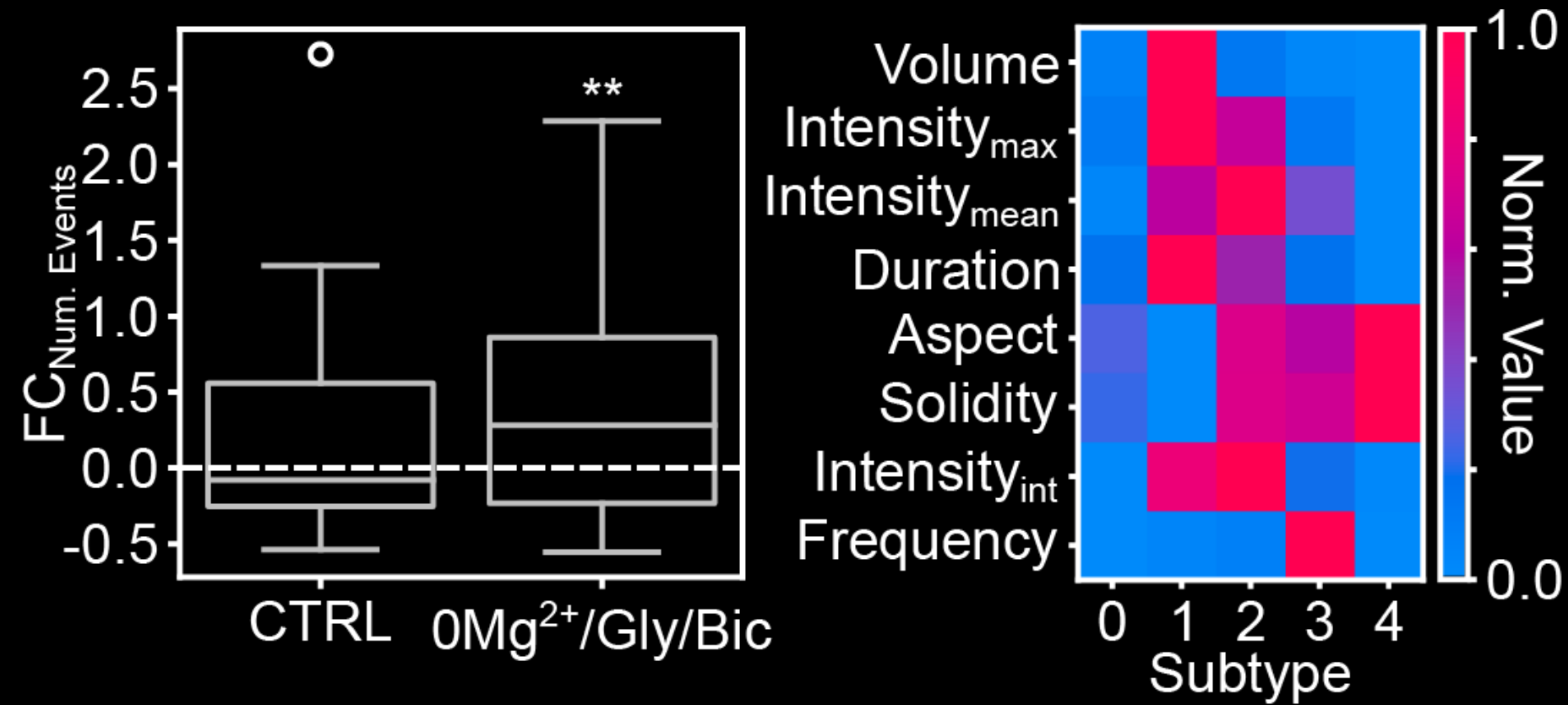
After cLTP



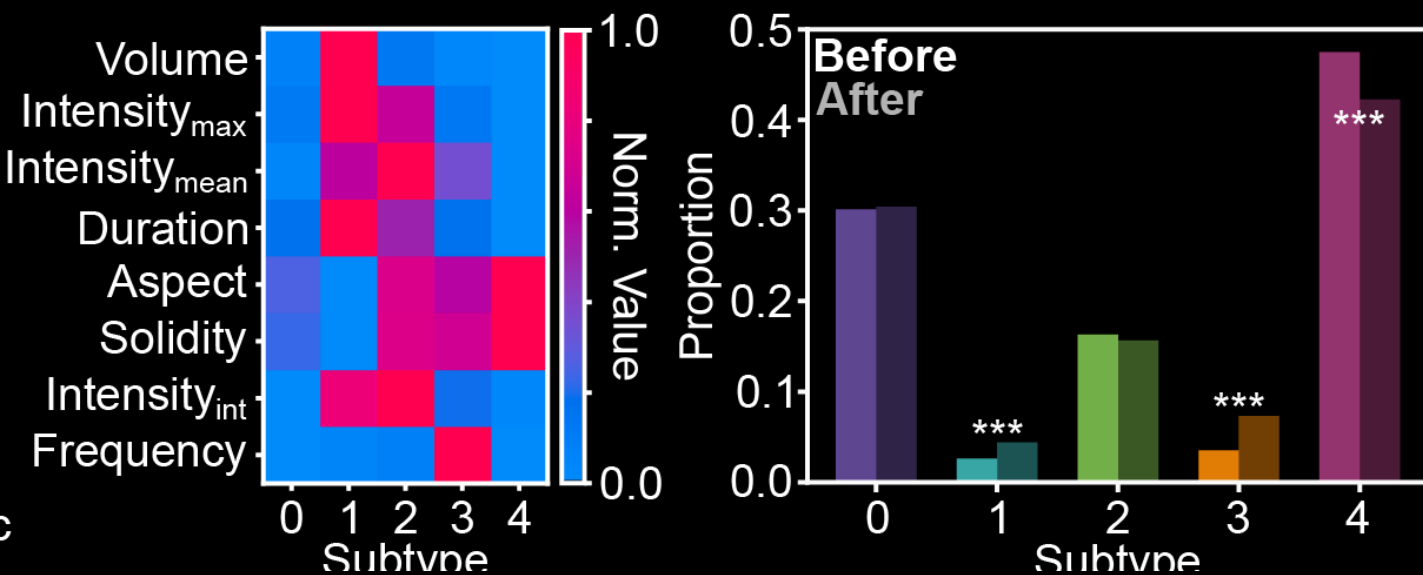
Synaptic potentiation is supported by an increase in the number of detected calcium transients



Synaptic potentiation is supported by a larger number of events



Only some mSCTs subtypes are affected by cLTP



c

